CUSUM Active-Manager Monitor

Statistical Process Control for Portfolio Manager Oversight

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What is CUSUM?

CUSUM (Cumulative Sum) is a sequential change-point detection method from statistical process control, adapted for investment management by Philips, Yashchin & Stein (2003). It answers one critical question:

Has a fund manager lost their investment skill — and if so, exactly when?

The Case for CUSUM — Why CIOs Should Care

Cost of Inaction:

Retaining an underperforming manager for 3 extra years on a 500M mandate at IR=-0.3, TE=6% costs ~9M in excess losses. Early, systematic detection pays.

Faster than Rolling IR:

A trailing 3-year IR review needs 10+ years of data to distinguish skill from luck at 95% confidence. CUSUM detects persistent CHANGES in IR within 2-3 years.

Rigorous False-Alarm Control:

Threshold h is calibrated from the paper's ARL table: h=19.81 gives ~1 false alarm per 60 months (5 years) when the true IR=+0.5. Fewer spurious terminations.

Two-Sided Monitoring:

Detects skill LOSS (lower CUSUM) and unexpected skill GAIN (upper CUSUM), enabling both firing decisions and early-stage manager promotions.

Rolling Restart:

After each alarm L resets to zero — monitoring never stops. Multiple alarms build a richer signal; post-alarm returns can be tracked to validate each flag.

Industry Precedent:

SPC methods are standard in manufacturing, clinical trials, and nuclear operations. Mercer and WTW publish SPC-adjacent monitoring frameworks for manager oversight.

Actionable, Not Automatic:

An alarm triggers a structured review — not automatic termination. It forces the investment committee to ask: Is the alpha source still intact?

Reference: Philips, T., Yashchin, E. & Stein, D. (2003). "Using Statistical Process Control to Monitor Active Managers." Journal of Portfolio Management, 30(1), 86-94.

How CUSUM Works

Step-by-step mechanics of the Philips-Yashchin-Stein procedure

1

Step 1 — Log Excess Return

```
Each month:   et = ln[(1 + r_fund) / (1 + r_benchmark)]
```

Log returns are additive and symmetric — they handle compounding correctly.

2

Step 2 — EWMA Tracking-Error Estimate

```
sigma^2_t = gamma * sigma^2_{t-1} + (1-gamma) * 0.5*(et - e_{t-1})^2   [gamma=0.90]
```

Von Neumann adjacent-difference estimator: robust to drift in the mean, adapts as the manager's risk budget changes.

3

Step 3 — Monthly Information Ratio Estimate

```
IR_hat_t = (et * sqrt(12)) / sigma_{t-1}
```

Annualised IR using lagged sigma to keep the estimator approximately unbiased.

4

Step 4 — Lindley Recursion (Page's CUSUM)

```
L_t = max(0, L_{t-1} + drift)    drift = 0.5*(IR_good + IR_bad) - IR_hat_t
```

Default: IR_good=0.5, IR_bad=0.0 -> drift = 0.25 - IR_hat. L grows when manager underperforms, resets to 0 when they outperform. Upper CUSUM L_up uses drift_up = IR_hat - 0.5*(IR_exceptional + IR_good).

5

Step 5 — Rolling Alarm Rule

```
ALARM if L_t >= h   [default h = 19.81]   then reset L_t = 0 and continue
```

After each alarm L resets to zero — monitoring never stops. All alarm times are recorded; post-alarm returns validate each signal.

Alarm Threshold Reference Table (Table 2, Philips et al. 2003)

Threshold (h)	ARL — Good (IR=0.5)	ARL — Flat (IR=0)	ARL — Bad (IR=-0.5)
11.81	24.00	16.00	11.00
15.00	36.00	22.00	15.00
17.60	48.00	27.00	18.00
19.81	**60.00**	**32.00**	**21.00** *<- default*
21.79	72.00	37.00	23.00
23.59	84.00	41.00	25.00

Fund Universe

68 active managers — page 1 of 4

#	Ticker	Fund Name	Benchmark	Asset Class	Start	h
1	FCNTX	Fidelity Contrafund vs Russell 1000 Growth	IWF	US Large Growth	Jan 2005	19.81
2	DODGX	Dodge & Cox Stock vs Russell 1000 Value	IWD	US Large Value	Jan 2005	19.81
3	AGTHX	AmFunds Growth Fund of America vs S&P 500	SPY	US Large Blend	Jan 2005	19.81
4	OTCFX	T. Rowe Price Small-Cap Stock vs Russell 2000	IWM	US Small Cap	Jan 2005	19.81
5	OAKIX	Oakmark International vs MSCI EAFE	EFA	Intl Developed	Jan 2005	19.81
6	TEDMX	Templeton Developing Markets vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
7	VWELX	Vanguard Wellington vs 60/40 Allocation	AOR	Balanced 60/40	Jan 2005	19.81
8	PTTRX	PIMCO Total Return vs US Agg Bond	AGG	US Fixed Income	Jan 2005	19.81
9	PRWCX	T. Rowe Price Capital Appreciation vs 60/40 Allocation	AOR	Balanced 60/40	Jan 2005	19.81
10	FMAGX	Fidelity Magellan vs S&P 500	SPY	US Large Blend	Jan 2005	19.81
11	SEQUX	Sequoia Fund vs S&P 500	SPY	US Large Blend	Jan 2005	19.81
12	FLPSX	Fidelity Low-Priced Stock vs Russell 2000	IWM	US Small/Mid Cap	Jan 2005	19.81
13	DODWX	Dodge & Cox International vs MSCI EAFE	EFA	Intl Value	Jan 2005	19.81
14	LSBRX	Loomis Sayles Bond vs US Agg Bond	AGG	US Fixed Income	Jan 2005	19.81
15	NWFFX	New World Fund vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
16	PRMSX	T. Rowe Price EM Stock vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
17	JEMWX	JPMorgan EM Equity vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
18	ODMAX	Invesco Developing Markets vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
19	PEIFX	PGIM Jennison EM Equity vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
20	LZOEX	Lazard EM Equity vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81

US Equity (FCNTX, DODGX, AGTHX, OTCFX, FMAGX, SEQUX, FLPSX)
Intl & Emerging Markets (OAKIX, TEDMX, DODWX, + 26 EM equity funds)
Fixed Income / Balanced (VWELX, PTTRX, PRWCX, LSBRX)
EM Debt: Hard CCY vs EMB (PEBIX, EBNDX, PREMX ...) & Local CCY vs EMLC
Global Equity vs ACWI / URTH (CWGIX, VHGEX, AIVSX ... + AIAIM funds)

Fund Universe

68 active managers — page 2 of 4

#	Ticker	Fund Name	Benchmark	Asset Class	Start	h
21	HLMEX	Harding Loevner EM Equity vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
22	MEMCX	Matthews EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
23	VEIEX	Vanguard EM Stock Index vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
24	VEMAX	Vanguard EM Stock Index Admiral vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
25	NEWFX	American Funds New World vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
26	HAEMX	Harbor EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
27	TEMFX	Templeton EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
28	SFENX	Seafarer EM Growth & Income vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
29	REMIX	Royce EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
30	ABEMX	AB EM Growth vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
31	GERIX	Goldman Sachs EM Equity vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
32	HLEMX	Harding Loevner EM Institutional vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
33	BGEFX	Baron EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
34	WIEMX	William Blair EM Growth vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
35	WAEMX	Western Asset EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
36	MIEMX	MFS EM Equity vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
37	EEMAX	Eaton Vance EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
38	FSEAX	Fidelity EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
39	MADCX	Madison EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81
40	PDEZX	PGIM EM vs MSCI EM	EEM	Emerging Markets	Jan 2005	19.81

Fund Universe

68 active managers — page 3 of 4

#	Ticker	Fund Name	Benchmark	Asset Class	Start	h
41	PEBIX	PIMCO EM Bond vs JPM EM Bond (Hard CCY)	EMB	EM Hard CCY Debt	Jan 2005	19.81
42	EBNDX	BlackRock EM Bond vs JPM EM Bond (Hard CCY)	EMB	EM Hard CCY Debt	Jan 2005	19.81
43	PREMX	T. Rowe Price EM Bond vs JPM EM Bond (Hard CCY)	EMB	EM Hard CCY Debt	Jan 2005	19.81
44	ABEMX	AB EM Debt vs JPM EM Bond (Hard CCY)	EMB	EM Hard CCY Debt	Jan 2005	19.81
45	TGBAX	Templeton Global Bond vs JPM EM Bond (Hard CCY)	EMB	EM Hard CCY Debt	Jan 2005	19.81
46	IEMIX	Invesco EM Sovereign Debt vs JPM EM Bond (Hard CCY)	EMB	EM Hard CCY Debt	Jan 2005	19.81
47	DBLEX	DFA EM Fixed Income vs JPM EM Bond (Hard CCY)	EMB	EM Hard CCY Debt	Jan 2005	19.81
48	VWOB	Vanguard EM Govt Bond ETF vs JPM EM Bond (Hard CCY)	EMB	EM Hard CCY Debt	Jan 2005	19.81
49	MSEDX	MFS EM Debt vs JPM EM Bond (Hard CCY)	EMB	EM Hard CCY Debt	Jan 2005	19.81
50	PFMIX	PIMCO Foreign Bond vs VanEck EM Local CCY Bond	EMLC	EM Local CCY Debt	Jan 2005	19.81
51	PRLAX	T. Rowe Price EM Local CCY Bond vs VanEck EM Local CCY Bond	EMLC	EM Local CCY Debt	Jan 2005	19.81
52	CWGIX	Capital World Growth & Income vs MSCI ACWI	ACWI	Global Equity	Jan 2005	19.81
53	TEDIX	Templeton Global Equity vs MSCI ACWI	ACWI	Global Equity	Jan 2005	19.81
54	VHGEX	Vanguard Global Equity vs MSCI ACWI	ACWI	Global Equity	Jan 2005	19.81
55	OPGIX	Invesco Oppenheimer Global vs MSCI ACWI	ACWI	Global Equity	Jan 2005	19.81
56	GPROX	Goldman Sachs Global Equity vs MSCI ACWI	ACWI	Global Equity	Jan 2005	19.81
57	SGIGX	Schroders Global vs MSCI ACWI	ACWI	Global Equity	Jan 2005	19.81
58	MGGIX	MFS Global Equity vs MSCI ACWI	ACWI	Global Equity	Jan 2005	19.81
59	AIVSX	American Investors vs MSCI World	URTH	Global Equity	Jan 2005	19.81
60	MGIAX	Morgan Stanley Institutional Global vs MSCI World	URTH	Global Equity	Jan 2005	19.81

Fund Universe

68 active managers — page 4 of 4

#	Ticker	Fund Name	Benchmark	Asset Class	Start	h
61	JGVAX	JPMorgan Global Focus vs MSCI World	URTH	Global Equity	Jan 2005	19.81
62	HAINX	Harbor International vs MSCI World	URTH	Global Equity	Jan 2005	19.81
63	0P0001I6KI.SI	AIAIM Global Equity Fund A vs MSCI ACWI	ACWI	Global Equity	Jan 2005	19.81
64	0P00008T6G.SI	AIAIM Global Equity Fund C vs MSCI ACWI	ACWI	Global Equity	Jan 2005	19.81
65	0P00008T6D.SI	AIAIM Global Equity Fund D vs MSCI ACWI	ACWI	Global Equity	Jan 2005	19.81
66	0P0001RMIN.SI	AIAIM Global Equity Fund E vs MSCI ACWI	ACWI	Global Equity	Jan 2005	19.81
67	0P0001KL95.SI	AIAIM EM Bond Fund A vs JPM EM Bond	EMB	EM Hard CCY Debt	Jan 2005	19.81
68	0P00008T6N.SI	AIAIM EM Bond Fund B vs JPM EM Bond	EMB	EM Hard CCY Debt	Jan 2005	19.81

Summary Results

Risk-adjusted statistics (Jan 2005 — present, 24m calibration excluded) — page 1 of 4

Fund	Bench	IR (95% CI)	t-stat	Hit%	Up/Dn Cap	Cum Exc	#Al	Alarm?	First Alarm	Regime
FCNTX	IWF	+0.04 [-0.35,+0.41]	+0.17	47%	93%/86%	+3.4%	6	YES	Jan 2011	
DODGX	IWD	+0.14 [-0.29,+0.60]	+0.66	52%	105%/103%	+14.5%	5	YES	Sep 2008	GFC
AGTHX	SPY	+0.10 [-0.37,+0.60]	+0.45	53%	103%/101%	+9.9%	7	YES	Nov 2008	GFC
OTCFX	IWM	+0.32 [-0.14,+0.82]	+1.49	54%	92%/84%	+32.7%	4	YES	Jul 2008	GFC
OAKIX	EFA	+0.06 [-0.37,+0.51]	+0.27	49%	111%/110%	+11.0%	7	YES	Oct 2007	GFC
TEDMX	EEM	+0.02 [-0.32,+0.40]	+0.10	54%	96%/94%	+2.7%	4	YES	Mar 2009	GFC
VWELX	AOR	**+0.64 [+0.39,+1.10]**	**+2.66**	60%	**106%/85%**	+58.0%	0	NO	-	
PTTRX	AGG	**+0.50 [+0.04,+1.02]**	**+2.30**	64%	**107%/90%**	+22.2%	2	YES	Sep 2015	
PRWCX	AOR	**+0.97 [+0.66,+1.45]**	**+4.07**	66%	**111%/74%**	+119.1%	1	YES	Jan 2026	
FMAGX	SPY	**-0.22 [-0.71,+0.33]**	**-1.01**	49%	104%/113%	-24.0%	8	YES	Oct 2008	GFC
SEQUX	SPY	**-0.22 [-0.65,+0.23]**	**-1.03**	46%	85%/86%	-33.2%	6	YES	Jul 2014	
FLPSX	IWM	+0.09 [-0.29,+0.46]	+0.41	50%	77%/67%	+17.2%	5	YES	Sep 2008	GFC
DODWX	EFA	**+0.42 [-0.01,+0.92]**	**+1.79**	56%	**109%/95%**	+72.3%	2	YES	Jan 2016	
LSBRX	AGG	+0.22 [-0.29,+0.86]	+1.02	59%	**113%/80%**	+39.1%	3	YES	Jul 2008	GFC
NWFFX	EEM	+0.16 [-0.19,+0.52]	+0.73	51%	79%/68%	+29.7%	4	YES	Mar 2011	
PRMSX	EEM	**-0.10 [-0.59,+0.40]**	**-0.46**	50%	99%/99%	-9.6%	6	YES	Oct 2008	GFC
JEMWX	EEM	+0.39 [-0.24,+1.12]	+1.39	62%	**105%/97%**	+27.3%	2	YES	Jan 2022	
ODMAX	EEM	+0.11 [-0.29,+0.48]	+0.49	49%	91%/86%	+14.2%	5	YES	Apr 2015	

Summary Results

Risk-adjusted statistics (Jan 2005 — present, 24m calibration excluded) — page 2 of 4

Fund	Bench	IR (95% CI)	t-stat	Hit%	Up/Dn Cap	Cum Exc	#Al	Alarm?	First Alarm	Regime
PEIFX	EEM	+0.22 [-0.50,+0.90]	+0.73	54%	97%/85%	+22.9%	3	YES	Feb 2020	COVID
LZOEX	EEM	+0.15 [-0.29,+0.58]	+0.71	50%	94%/88%	+23.4%	4	YES	Sep 2015	
HLMEX	EEM	-0.06 [-0.43,+0.33]	-0.29	50%	93%/92%	-7.1%	5	YES	Dec 2008	GFC
MEMCX	EEM	-0.32 [-0.69,+0.01]	-1.49	50%	90%/93%	-26.9%	6	YES	Sep 2008	GFC
VEIEX	EEM	-0.08 [-0.42,+0.32]	-0.38	52%	93%/92%	-6.8%	4	YES	Oct 2008	GFC
VEMAX	EEM	-0.02 [-0.40,+0.37]	-0.09	54%	94%/92%	-1.5%	3	YES	Jun 2017	
NEWFX	EEM	+0.16 [-0.19,+0.52]	+0.73	51%	79%/68%	+29.4%	4	YES	Mar 2011	
HAEMX*	EEM	-0.55 [-1.06,+0.95]	-1.09	55%	61%/103%	-39.3%	0	NO	-	
TEMFX	EEM	-0.18 [-0.63,+0.24]	-0.84	48%	73%/73%	-33.1%	6	YES	Sep 2007	
SFENX	EEM	+0.08 [-0.39,+0.55]	+0.34	49%	95%/91%	+10.0%	5	YES	Jun 2012	
REMIX	EEM	+0.11 [-0.66,+0.91]	+0.28	43%	42%/4%	+13.1%	2	YES	Mar 2025	
ABEMX	EEM	+0.24 [-0.21,+0.69]	+1.05	55%	95%/88%	+30.8%	5	YES	Dec 2013	
GERIX	EEM	+0.23 [-0.20,+0.65]	+0.99	55%	101%/97%	+21.5%	2	YES	Sep 2019	
HLEMX	EEM	-0.42 [-1.04,+0.34]	-1.10	49%	100%/111%	-15.7%	2	YES	Dec 2023	
BGEFX	EEM	+0.05 [-0.62,+0.76]	+0.17	48%	99%/93%	+6.9%	2	YES	Feb 2022	
WIEMX	EEM	-0.09 [-0.68,+0.55]	-0.28	50%	87%/83%	-10.2%	3	YES	Jun 2022	
WAEMX	EEM	+0.16 [-0.31,+0.63]	+0.70	56%	88%/77%	+40.9%	5	YES	Dec 2013	
MIEMX	EEM	+0.02 [-0.45,+0.51]	+0.09	51%	84%/78%	+2.3%	4	YES	Nov 2011	

Summary Results

Risk-adjusted statistics (Jan 2005 — present, 24m calibration excluded) — page 3 of 4

Fund	Bench	IR (95% CI)	t-stat	Hit%	Up/Dn Cap	Cum Exc	#Al	Alarm?	First Alarm	Regime
EEMAX	EEM	+0.04 [-0.45,+0.55]	+0.19	51%	98%/96%	+4.7%	5	YES	Jan 2017	
FSEAX	EEM	**+0.42 [-0.07,+0.93]**	**+1.93**	59%	**101%/86%**	+120.3%	3	YES	May 2008	GFC
MADCX	EEM	+0.12 [-0.23,+0.47]	+0.57	51%	98%/95%	+14.7%	4	YES	Oct 2008	GFC
PDEZX	EEM	+0.19 [-0.44,+0.84]	+0.65	51%	**102%/89%**	+27.6%	2	YES	Apr 2022	
PEBIX	EMB	+0.11 [-0.20,+0.57]	+0.47	51%	91%/83%	+9.1%	2	YES	Oct 2014	
EBNDX	EMB	**-0.16 [-0.74,+0.49]**	**-0.53**	58%	87%/92%	-28.9%	2	YES	Sep 2011	
PREMX	EMB	**-0.01 [-0.34,+0.45]**	**-0.03**	53%	90%/87%	-0.5%	3	YES	Sep 2014	
ABEMX	EMB	+0.07 [-0.44,+0.60]	+0.31	51%	151%/154%	+20.5%	5	YES	Jan 2014	
TGBAX	EMB	**-0.17 [-0.65,+0.21]**	**-0.71**	46%	58%/56%	-25.7%	7	YES	Dec 2011	
IEMIX*	EMB	+1.58 [-0.79,+6.78]	+0.91	75%	-	+17.4%	0	NO	-	
DBLEX	EMB	**-0.03 [-0.48,+0.39]**	**-0.13**	48%	72%/61%	-2.3%	3	YES	Nov 2015	
VWOB	EMB	+0.08 [-0.37,+0.48]	+0.30	48%	94%/91%	+1.5%	3	YES	Jul 2016	
MSEDX*	EMB	**-1.41 [-2.34,-0.60]**	**-2.58**	35%	49%/93%	-18.6%	1	YES	Nov 2021	
PFMIX	EMLC	+0.26 [-0.18,+0.66]	+1.05	49%	32%/9%	+51.4%	2	YES	Jul 2017	
PRLAX	EMLC	+0.04 [-0.39,+0.49]	+0.16	49%	174%/160%	+12.1%	3	YES	Feb 2014	
CWGIX	ACWI	**-0.09 [-0.52,+0.31]**	**-0.40**	46%	95%/94%	-5.0%	5	YES	Oct 2016	
TEDIX	ACWI	**-0.16 [-0.74,+0.39]**	**-0.68**	46%	75%/72%	-22.3%	7	YES	Oct 2015	
VHGEX	ACWI	**-0.14 [-0.61,+0.35]**	**-0.59**	51%	102%/104%	-7.8%	4	YES	Dec 2021	

Summary Results

Risk-adjusted statistics (Jan 2005 — present, 24m calibration excluded) — page 4 of 4

Fund	Bench	IR (95% CI)	t-stat	Hit%	Up/Dn Cap	Cum Exc	#Al	Alarm?	First Alarm	Regime
OPGIX	ACWI	-0.09 [-0.57,+0.42]	-0.36	48%	112%/118%	-16.6%	7	YES	Jul 2012	
GPROX	ACWI	-0.35 [-0.98,+0.30]	-1.27	51%	96%/108%	-27.8%	5	YES	Aug 2019	
SGIGX	ACWI	-0.42 [-1.18,+0.33]	-1.21	44%	89%/94%	-15.9%	4	YES	Jun 2020	
MGGIX	ACWI	+0.38 [-0.29,+1.09]	+1.62	59%	114%/96%	+94.9%	4	YES	Oct 2012	
AIVSX	URTH	+0.50 [+0.03,+0.97]	+1.88	55%	99%/86%	+28.5%	2	YES	Jul 2019	
MGIAX	URTH	-0.18 [-0.65,+0.23]	-0.67	44%	85%/81%	-16.7%	4	YES	Sep 2018	
JGVAX*	URTH	-1.11 [-3.20,-0.89]	-1.67	26%	11%/100%	-55.9%	0	NO	-	
HAINX	URTH	-0.74 [-1.32,-0.26]	-2.79	44%	83%/103%	-50.9%	7	YES	Oct 2014	
0P0001I6KIA*	ACWI	-1.15 [-2.00,-0.39]	-2.26	37%	64%/69%	-24.0%	1	YES	Jun 2025	
0P00008T6GA*	ACWI	-0.53 [-1.48,+0.26]	-1.23	39%	61%/61%	-24.9%	1	YES	Nov 2023	
0P00008T6DA*	ACWI	+0.64 [-0.20,+1.61]	+1.49	55%	126%/71%	+59.2%	0	NO	-	
0P0001RMIN*	ACWI*	-1.46 [-2.86,-0.35]	-2.35	32%	45%/41%	-26.6%	0	NO	-	
0P0001KL9E*	EMB*	-0.07 [-1.15,+0.99]	-0.12	41%	63%/23%	-1.3%	1	YES	Feb 2026	
0P00008T6NE*	EMB	-0.53 [-1.24,+0.11]	-1.23	44%	39%/19%	-14.3%	2	YES	Aug 2024	

Multiple-Testing Caution | Key Observations

MULTIPLE TESTING: With 68 funds at h=19.81 (ARL=60m per fund), expected FALSE alarms = 13.6/year across the universe. GFC alarms: 14 | COVID alarms: 1 — these are regime signals, not skill-loss signals.
* = Short monitoring history (<36m live CUSUM; treat results cautiously).
62/68 alarmed. Clean: VWELX, HAEMX, IEMIX, JGVAX.... Best IR: IEMIX (+1.58, t=+0.9). Most alarms: FMAGX (8x).

Post-Alarm Analysis

Cumulative excess return vs benchmark in the months following the FIRST alarm — page 1 of 5

Fund	First Alarm	+12M Fund	+12M Bench	+12M Exc	+24M Exc	+36M Exc	1st Sig	#Al	Sig%(all)
FCNTX	Jan 2011	+4.3%	+5.9%	-1.7%	-0.1%	+2.6%	CORRECT	6	67%
DODGX	Sep 2008	-3.7%	-10.5%	+6.9%	+5.5%	+3.3%	FALSE	5	0%
AGTHX	Nov 2008	+33.1%	+25.2%	+7.9%	+6.7%	-0.3%	FALSE	7	33%
OTCFX	Jul 2008	-12.8%	-20.8%	+8.0%	+14.7%	+24.0%	FALSE	4	33%
OAKIX	Oct 2007	-43.3%	-45.9%	+2.6%	+14.2%	+21.3%	FALSE	7	17%
TEDMX	Mar 2009	+83.6%	+72.6%	+11.0%	+11.1%	+14.4%	FALSE	4	25%
PTTRX	Sep 2015	+5.7%	+5.2%	+0.6%	+3.2%	+3.0%	FALSE	2	0%
PRWCX	Jan 2026	n/a	n/a	n/a	n/a	n/a	FALSE	1	n/a
FMAGX	Oct 2008	+19.3%	+9.7%	+9.6%	+8.5%	-4.3%	FALSE	8	33%
SEQUX	Jul 2014	+29.5%	+11.2%	+18.3%	-24.3%	-31.1%	FALSE	6	60%
FLPSX	Sep 2008	+4.8%	-9.9%	+14.7%	+17.9%	+22.9%	FALSE	5	0%
DODWX	Jan 2016	+31.9%	+10.8%	+21.1%	+22.8%	+23.0%	FALSE	2	0%
LSBRX	Jul 2008	-0.8%	+7.3%	-8.0%	+1.6%	+12.6%	CORRECT	3	67%
NWFFX	Mar 2011	-4.4%	-10.0%	+5.6%	+12.1%	+23.2%	FALSE	4	0%

Post-Alarm Analysis

Cumulative excess return vs benchmark in the months following the FIRST alarm — page 2 of 5

Fund	First Alarm	+12M Fund	+12M Bench	+12M Exc	+24M Exc	+36M Exc	1st Sig	#Al	Sig%(all)
PRMSX	Oct 2008	+60.4%	+51.0%	+9.4%	+13.2%	+13.6%	FALSE	6	60%
JEMWX	Jan 2022	-13.9%	-13.3%	-0.6%	-2.3%	-4.4%	CORRECT	2	50%
ODMAX	Apr 2015	-14.1%	-17.9%	+3.8%	+4.5%	+6.0%	FALSE	5	50%
PEIFX	Feb 2020	+27.4%	+34.7%	-7.4%	+12.5%	+23.3%	CORRECT	3	50%
LZOEX	Sep 2015	+23.3%	+16.9%	+6.4%	+1.9%	-6.8%	FALSE	4	50%
HLMEX	Dec 2008	+63.9%	+69.0%	-5.0%	+1.3%	+7.3%	CORRECT	5	50%
MEMCX	Sep 2008	+10.5%	+16.4%	-5.9%	-3.1%	+3.1%	CORRECT	6	80%
VEIEX	Oct 2008	+56.0%	+51.0%	+4.9%	+4.7%	+5.6%	FALSE	4	50%
VEMAX	Jun 2017	+6.3%	+7.0%	-0.7%	+1.3%	+2.7%	CORRECT	3	100%
NEWFX	Mar 2011	-4.4%	-10.0%	+5.6%	+12.1%	+23.2%	FALSE	4	0%
TEMFX	Sep 2007	-27.3%	-29.7%	+2.4%	-0.1%	-12.9%	FALSE	6	80%
SFENX	Jun 2012	+2.3%	+0.3%	+2.0%	+5.0%	-2.7%	FALSE	5	50%
REMIX	Mar 2025	+16.8%	+33.1%	-16.3%	n/a	n/a	CORRECT	2	100%
ABEMX	Dec 2013	-2.4%	-3.9%	+1.5%	+3.6%	+4.9%	FALSE	5	50%

Post-Alarm Analysis

Cumulative excess return vs benchmark in the months following the FIRST alarm — page 3 of 5

Fund	First Alarm	+12M Fund	+12M Bench	+12M Exc	+24M Exc	+36M Exc	1st Sig	#Al	Sig%(all)
GERIX	Sep 2019	+11.8%	+10.8%	+0.9%	+8.0%	+7.3%	FALSE	2	0%
HLEMX	Dec 2023	+2.0%	+6.5%	-4.5%	-10.8%	n/a	CORRECT	2	100%
BGEFX	Feb 2022	-14.0%	-16.2%	+2.2%	+4.8%	+7.2%	FALSE	2	0%
WIEMX	Jun 2022	+8.9%	+0.9%	+8.0%	+10.2%	-9.2%	FALSE	3	50%
WAEMX	Dec 2013	+0.9%	-3.9%	+4.8%	+10.9%	-1.3%	FALSE	5	25%
MIEMX	Nov 2011	+8.7%	+6.7%	+1.9%	-1.2%	+6.3%	FALSE	4	0%
EEMAX	Jan 2017	+46.8%	+39.4%	+7.5%	-1.3%	+16.8%	FALSE	5	20%
FSEAX	May 2008	-34.3%	-32.4%	-1.9%	-4.9%	+0.1%	CORRECT	3	67%
MADCX	Oct 2008	+68.3%	+51.0%	+17.3%	+15.9%	+13.7%	FALSE	4	50%
PDEZX	Apr 2022	-18.2%	-5.5%	-12.7%	-0.8%	+1.0%	CORRECT	2	100%
PEBIX	Oct 2014	-5.7%	-0.9%	-4.8%	-3.2%	-1.2%	CORRECT	2	50%
EBNDX	Sep 2011	+2.7%	+20.6%	-17.8%	+20.8%	-7.9%	CORRECT	2	100%
PREMX	Sep 2014	-3.6%	-1.4%	-2.2%	+1.1%	+4.2%	CORRECT	3	67%
ABEMX	Jan 2014	+8.0%	+8.9%	-0.9%	-20.2%	-12.5%	CORRECT	5	60%

Post-Alarm Analysis

Cumulative excess return vs benchmark in the months following the FIRST alarm — page 4 of 5

Fund	First Alarm	+12M Fund	+12M Bench	+12M Exc	+24M Exc	+36M Exc	1st Sig	#Al	Sig%(all)
TGBAX	Dec 2011	+16.2%	+16.5%	-0.3%	+11.6%	+7.2%	CORRECT	7	86%
DBLEX	Nov 2015	+10.8%	+6.0%	+4.7%	+4.0%	+6.8%	FALSE	3	0%
VWOB	Jul 2016	+4.2%	+3.9%	+0.3%	+1.0%	+1.7%	FALSE	3	0%
MSEDX	Nov 2021	n/a	n/a	n/a	n/a	n/a	FALSE	1	n/a
PFMIX	Jul 2017	+3.5%	-4.1%	+7.6%	+8.6%	+18.1%	FALSE	2	50%
PRLAX	Feb 2014	-5.3%	-5.3%	-0.1%	-13.4%	+8.7%	CORRECT	3	33%
CWGIX	Oct 2016	+23.2%	+23.8%	-0.5%	-4.7%	-4.8%	CORRECT	5	60%
TEDIX	Oct 2015	+4.0%	+2.5%	+1.6%	-7.9%	-10.7%	FALSE	7	67%
VHGEX	Dec 2021	-22.7%	-18.4%	-4.3%	-4.3%	-8.9%	CORRECT	4	100%
OPGIX	Jul 2012	+25.2%	+20.9%	+4.3%	-2.0%	+10.5%	FALSE	7	57%
GPROX	Aug 2019	+33.1%	+16.5%	+16.6%	+45.7%	+3.8%	FALSE	5	75%
SGIGX	Jun 2020	+42.3%	+39.3%	+2.9%	+13.4%	+1.4%	FALSE	4	67%
MGGIX	Oct 2012	+34.9%	+23.4%	+11.5%	+27.5%	+55.4%	FALSE	4	33%
AIVSX	Jul 2019	+8.9%	+7.3%	+1.7%	-1.3%	+0.8%	FALSE	2	0%

Post-Alarm Analysis

Cumulative excess return vs benchmark in the months following the FIRST alarm — page 5 of 5

Fund	First Alarm	+12M Fund	+12M Bench	+12M Exc	+24M Exc	+36M Exc	1st Sig	#Al	Sig%(all)
MGIAX	Sep 2018	+4.2%	+2.3%	**+1.9%**	**+9.2%**	**-7.1%**	**FALSE**	4	**25%**
HAINX	Oct 2014	-1.4%	+1.4%	**-2.8%**	**-8.4%**	**-15.1%**	**CORRECT**	7	**71%**
0P0001I6KI.SI	Jun 2025	n/a	n/a	**n/a**	**n/a**	**n/a**	**FALSE**	1	n/a
0P00008T6G.S	Nov 2023	+30.0%	+26.5%	**+3.5%**	**+6.4%**	**n/a**	**FALSE**	1	**0%**
0P0001KL95.SI	Feb 2026	n/a	n/a	**n/a**	**n/a**	**n/a**	**FALSE**	1	n/a
0P00008T6N.SI	Aug 2024	+8.9%	+8.0%	**+0.8%**	**n/a**	**n/a**	**FALSE**	2	**0%**

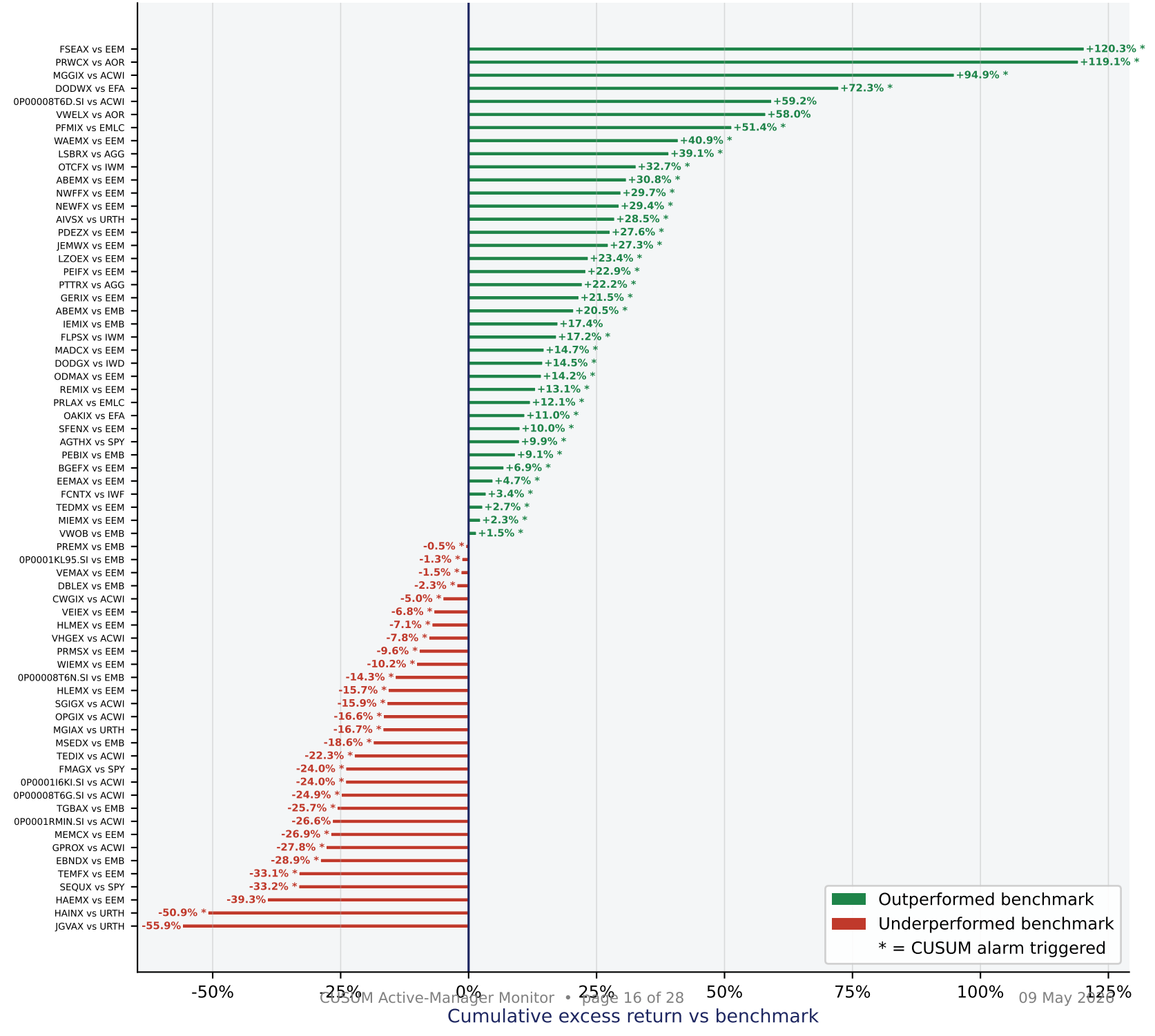
No alarm (no post-alarm analysis): VWELX, HAEMX, IEMIX, JGVAX, 0P00008T6D.SI, 0P0001RMIN.SI

How to Read This Table

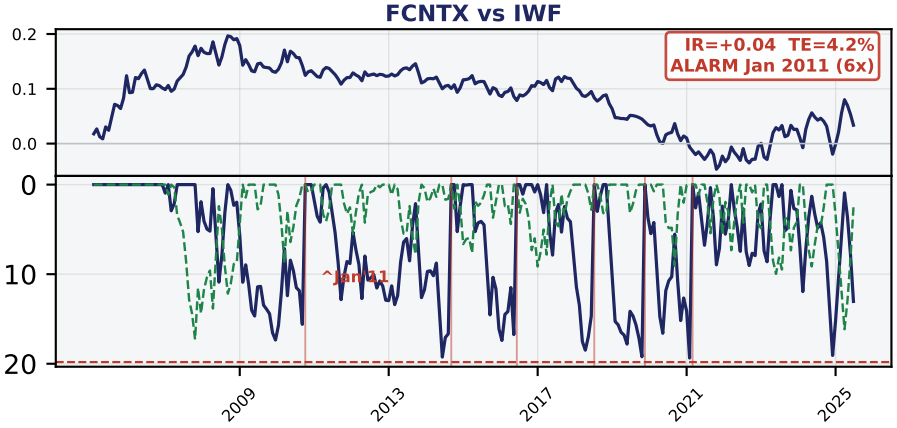
+12M/+24M/+36M: cumulative excess return (fund minus benchmark) in the months AFTER the FIRST alarm.
1st Sig: CORRECT if fund underperformed after first alarm (signal validated). FALSE = false positive.
Sig%(all): % of ALL alarms (rolling restart) where fund underperformed at +12M. >60% = reliable signal. 50% = coin flip. At h=19.81, false-alarm rate = 1 per 60 months PER FUND — across a 67-fund universe expect ~13 false alarms per year.

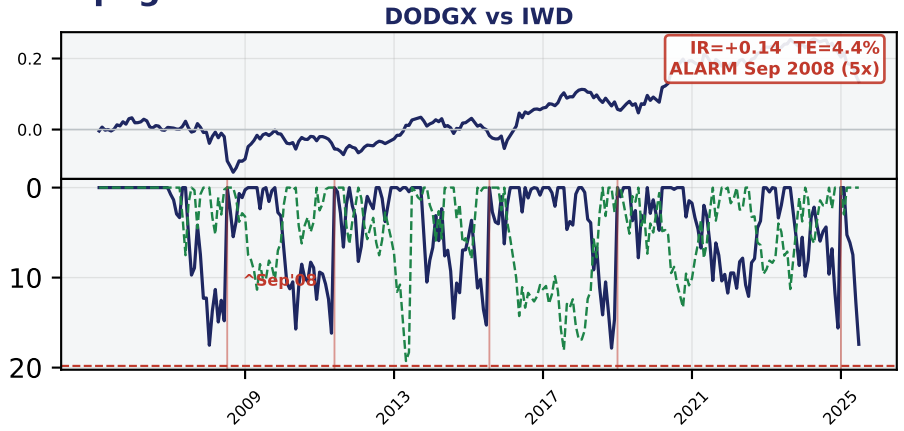
Cumulative Excess Returns — Ranked

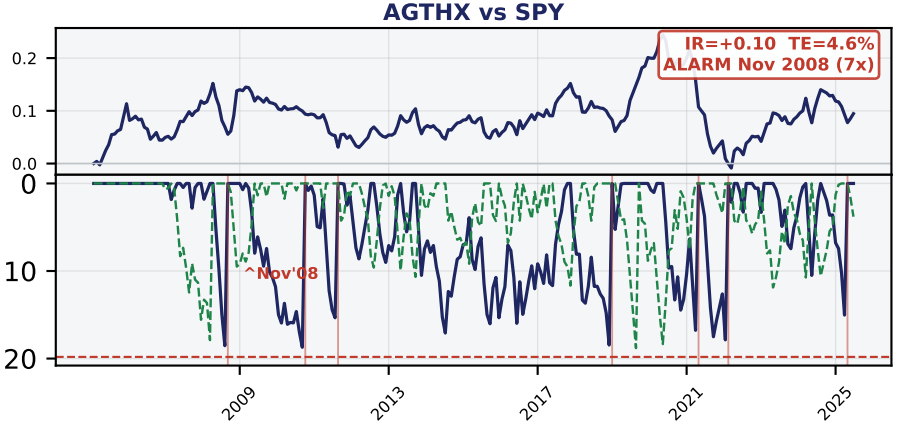
Total return of each fund vs its benchmark (Jan 2005 - present)

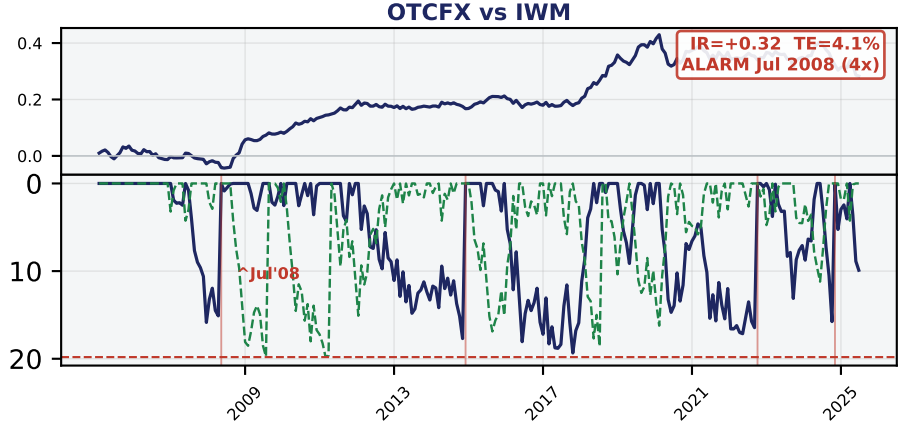


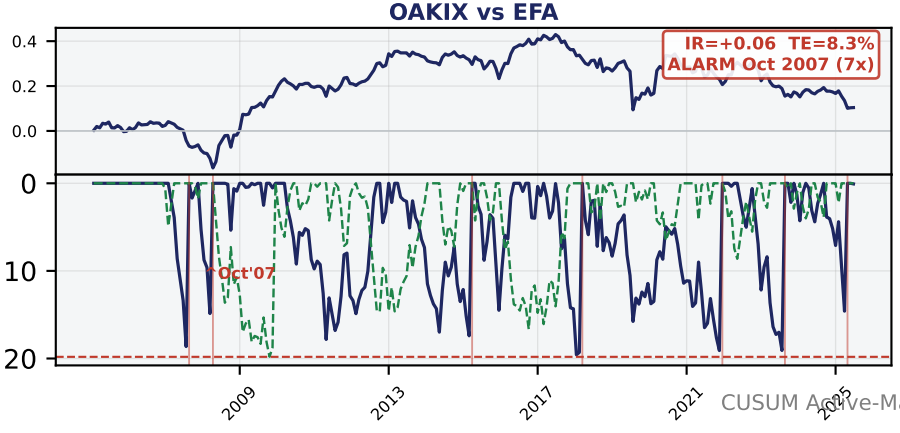
Individual Fund CUSUM Charts — page 1 of 12

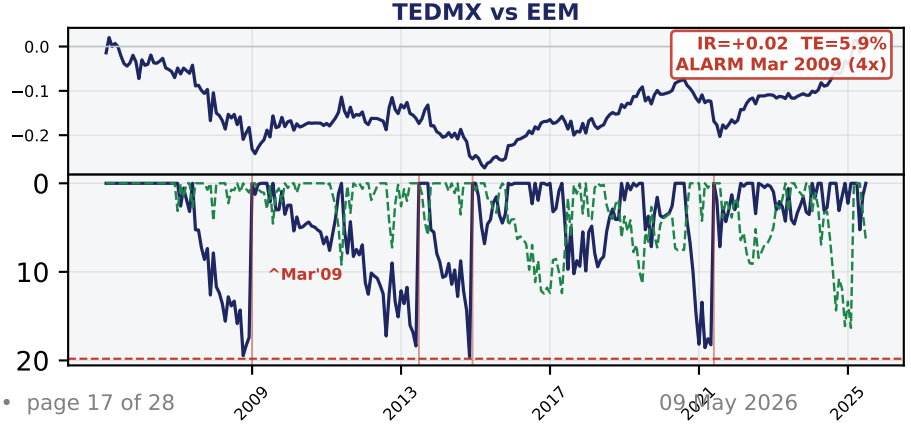




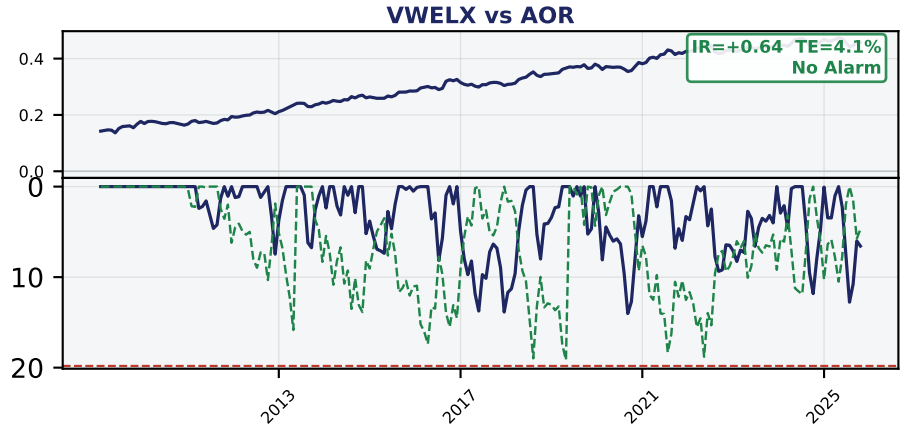


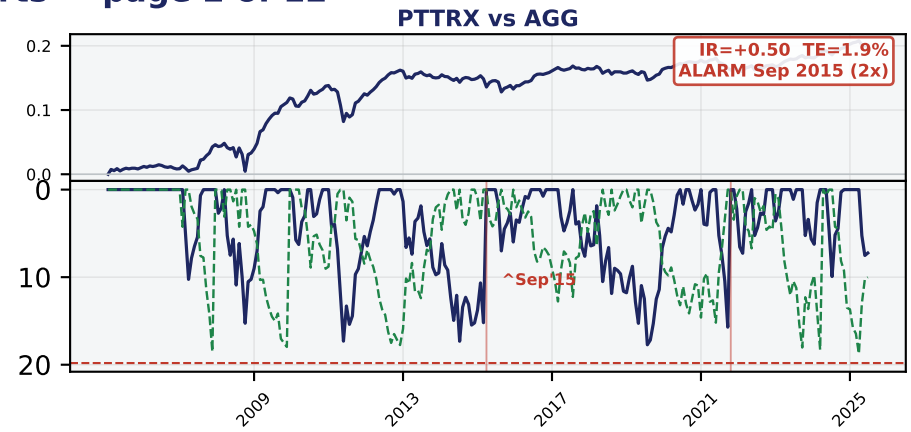


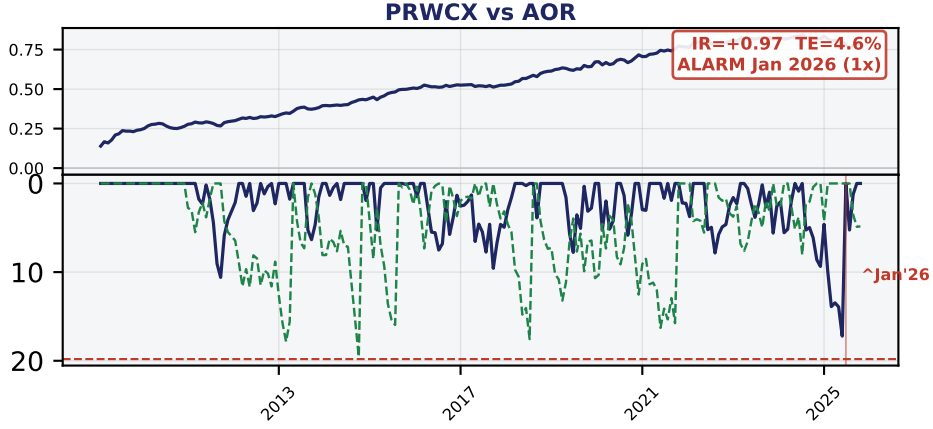


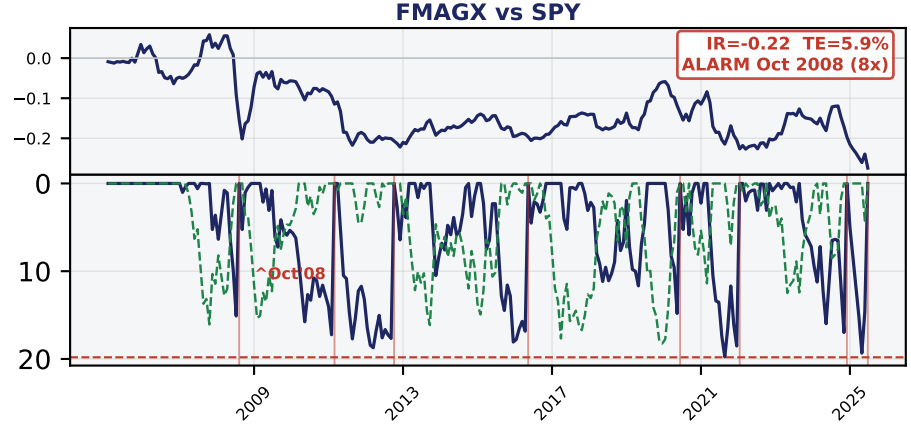


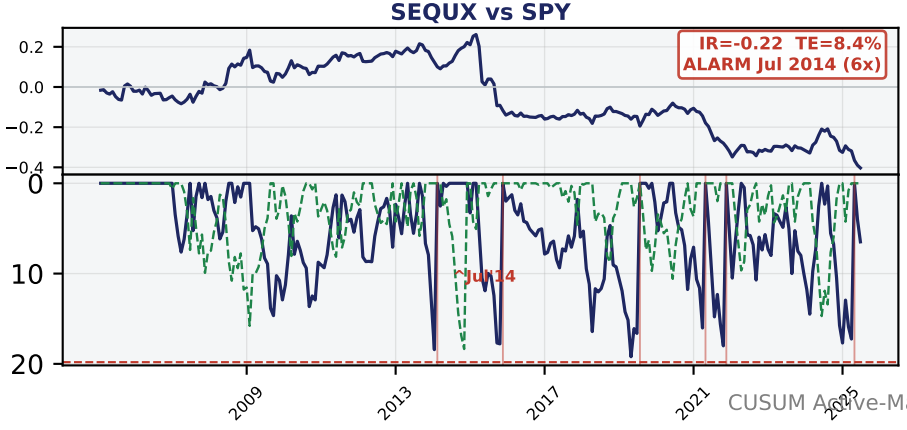
Individual Fund CUSUM Charts — page 2 of 12

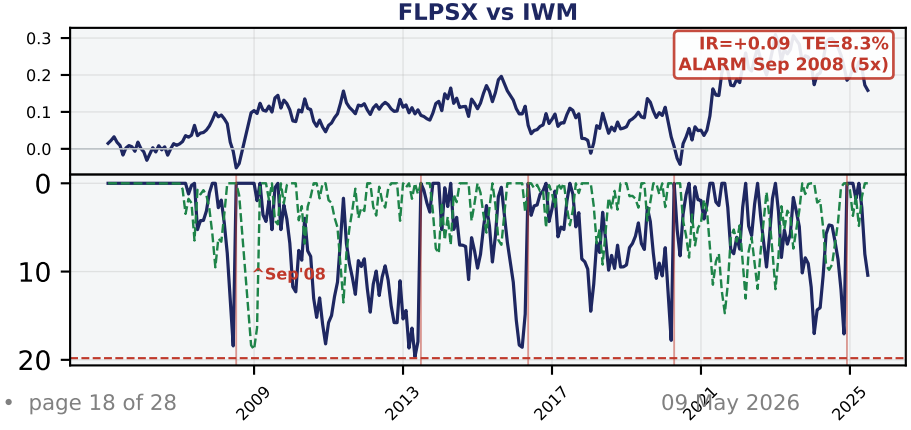




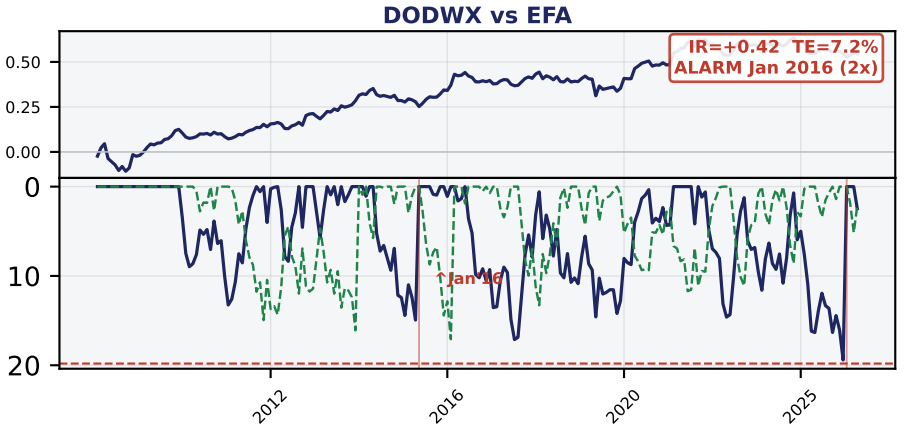


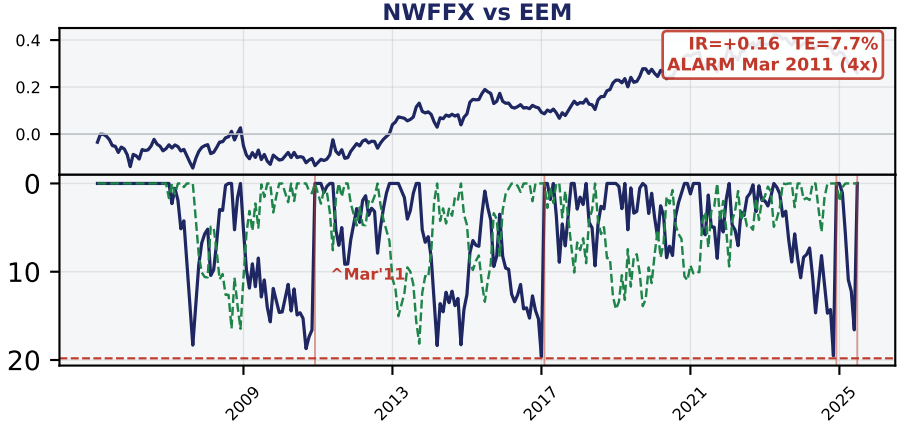


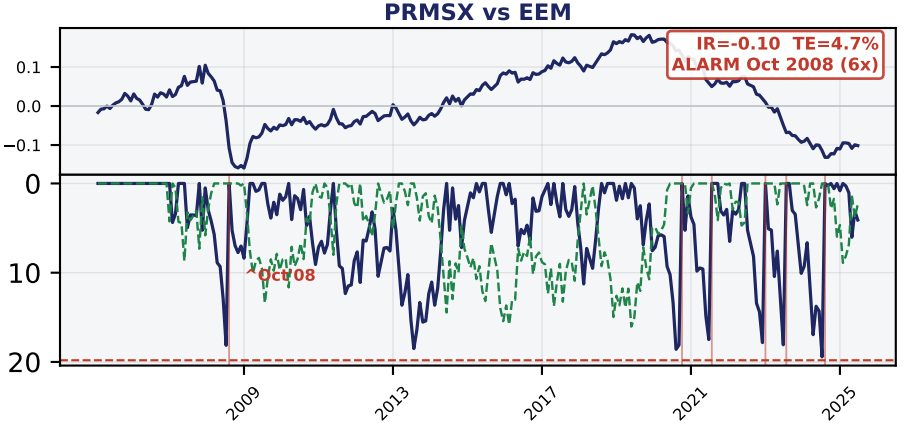


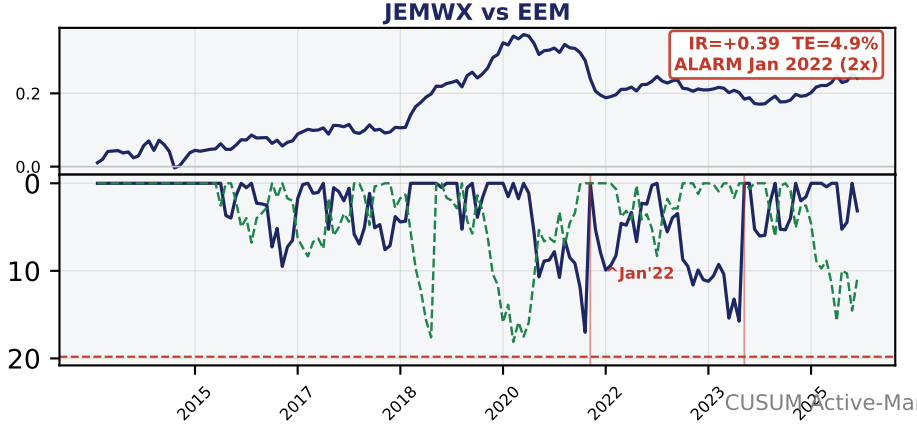


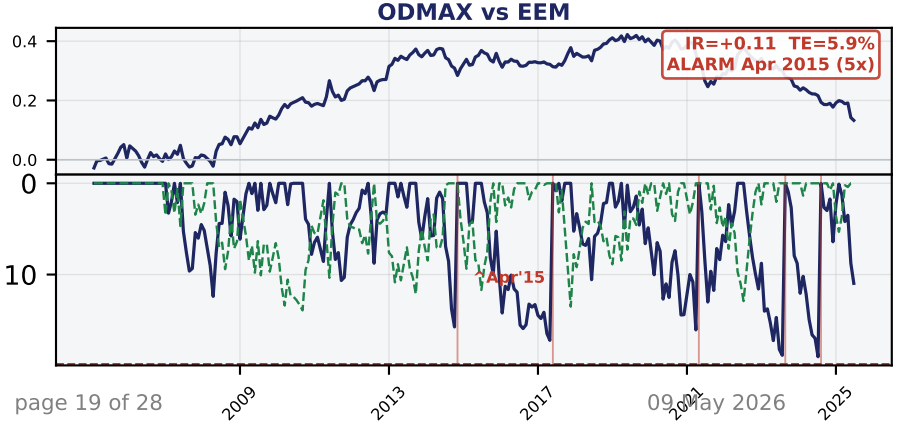
Individual Fund CUSUM Charts — page 3 of 12



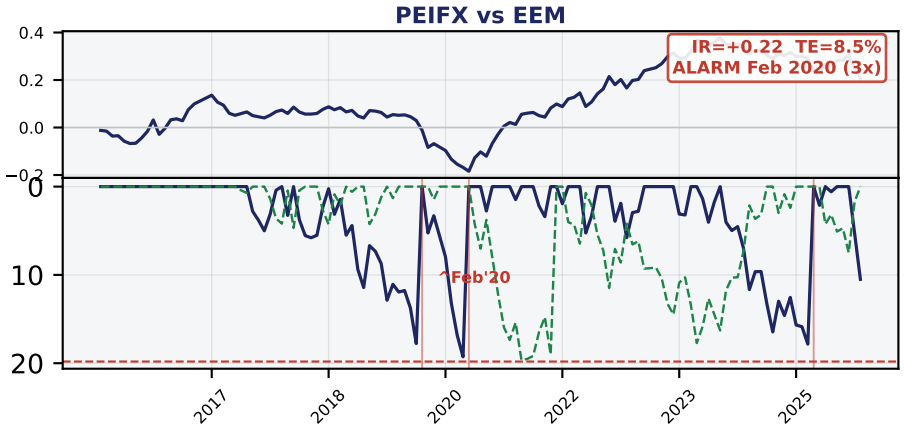


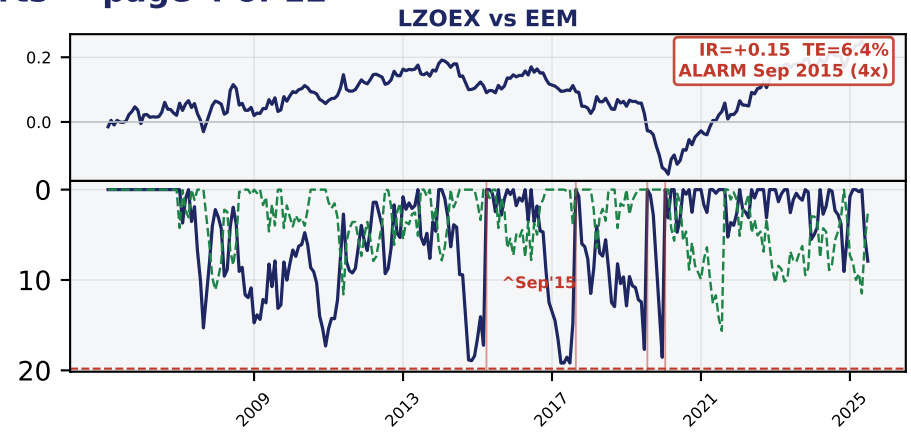


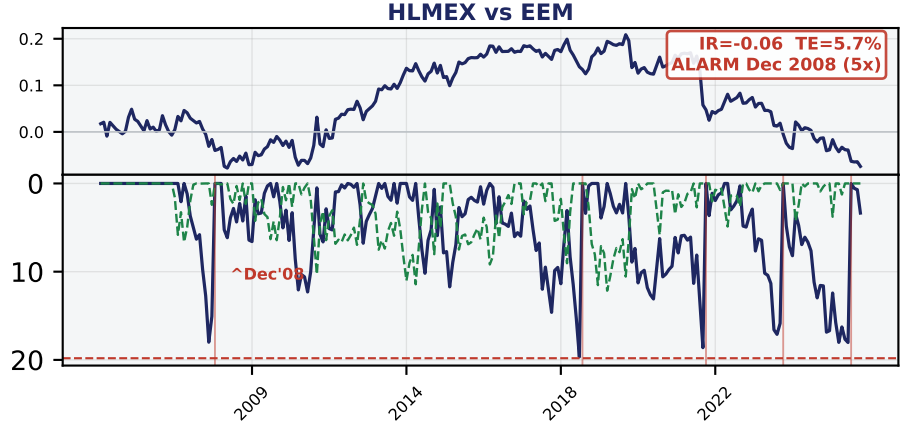


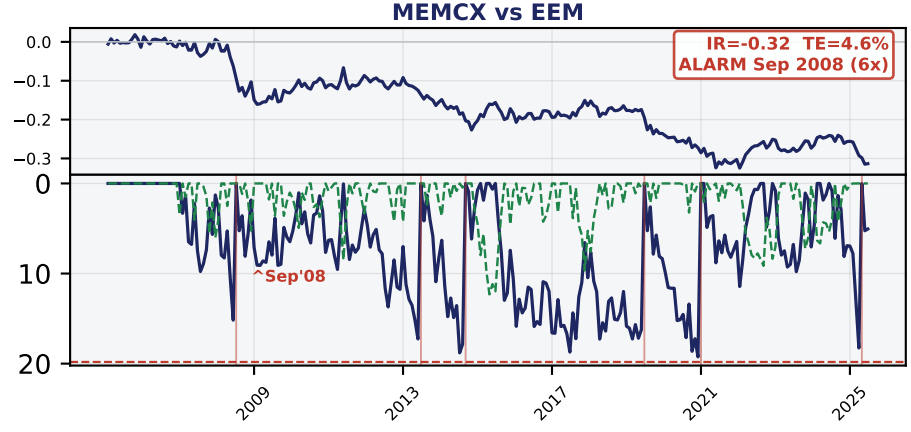


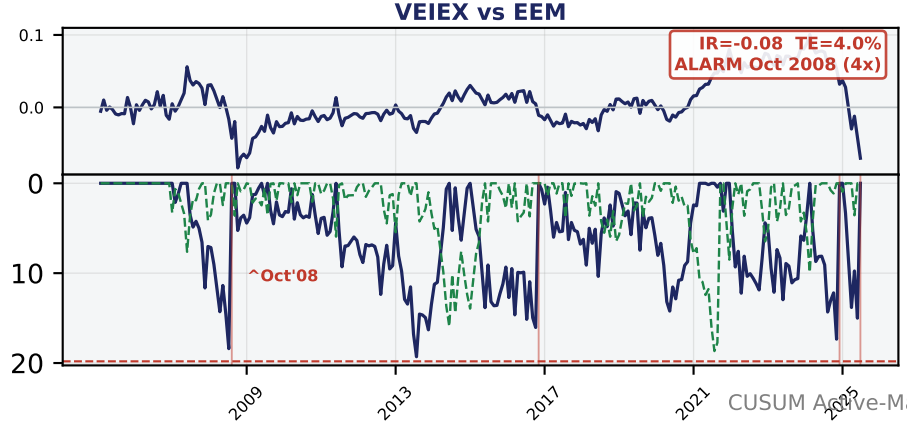
Individual Fund CUSUM Charts — page 4 of 12

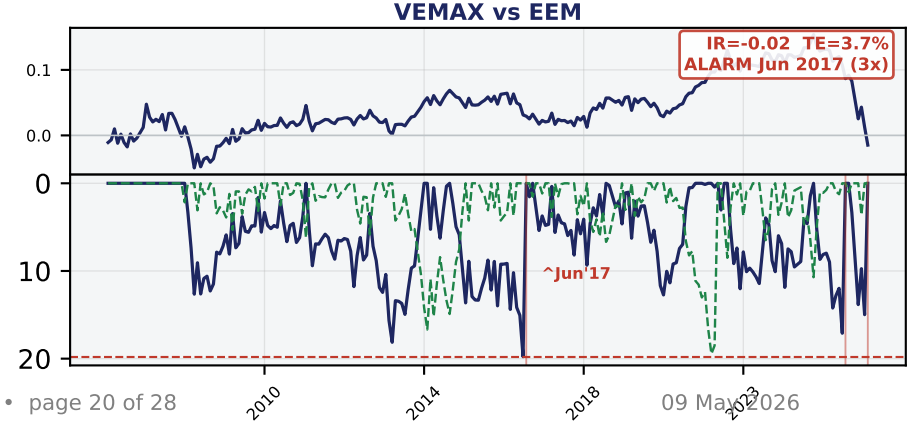




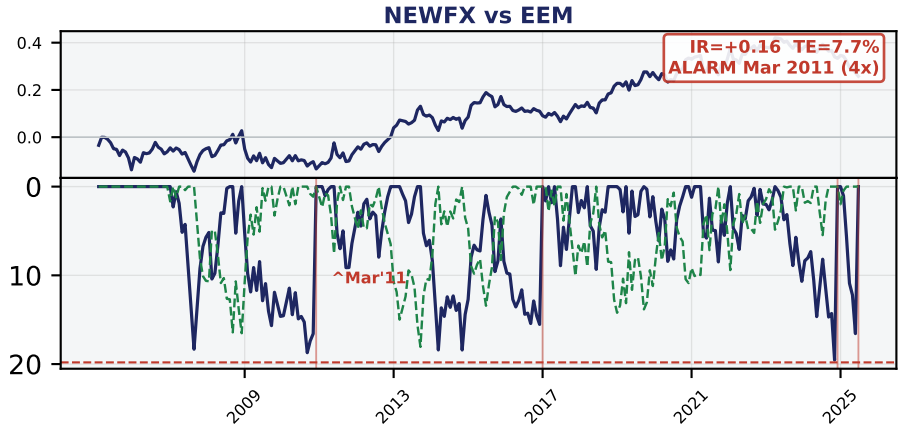


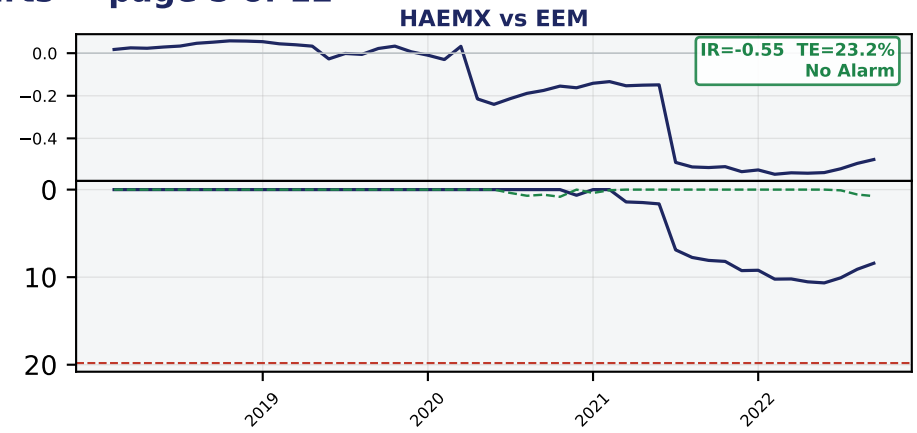


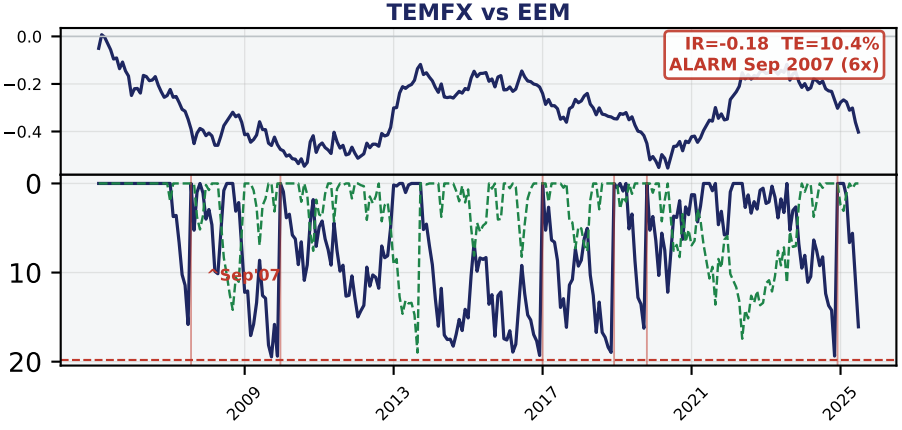


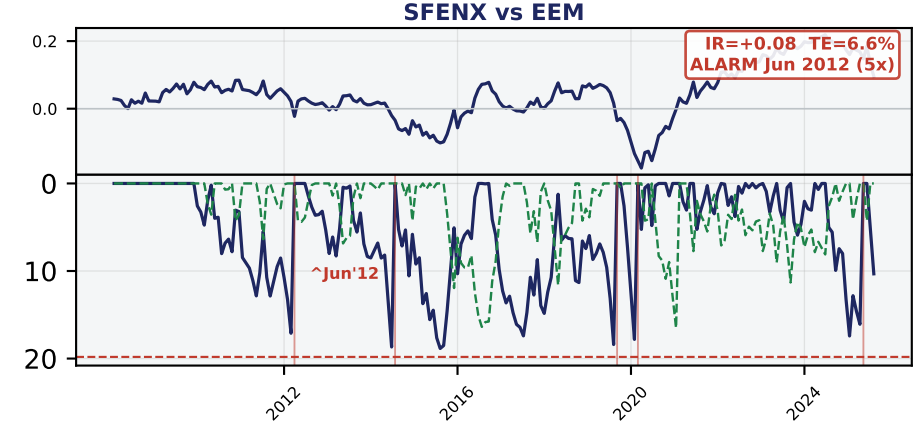


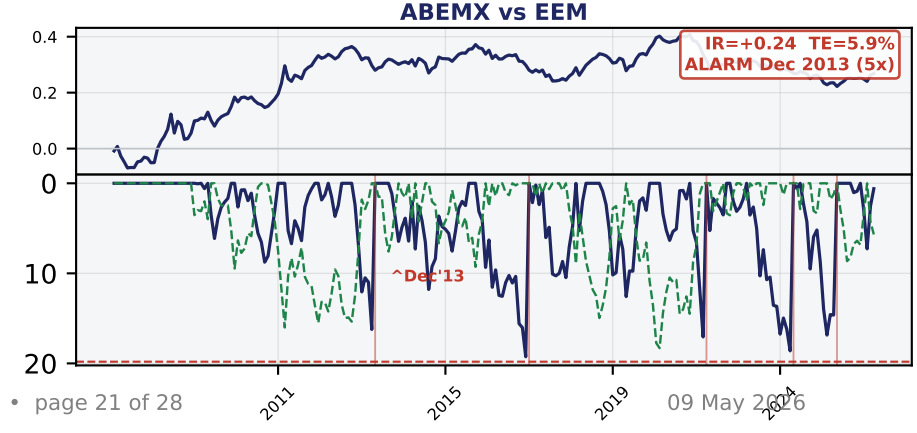
Individual Fund CUSUM Charts — page 5 of 12



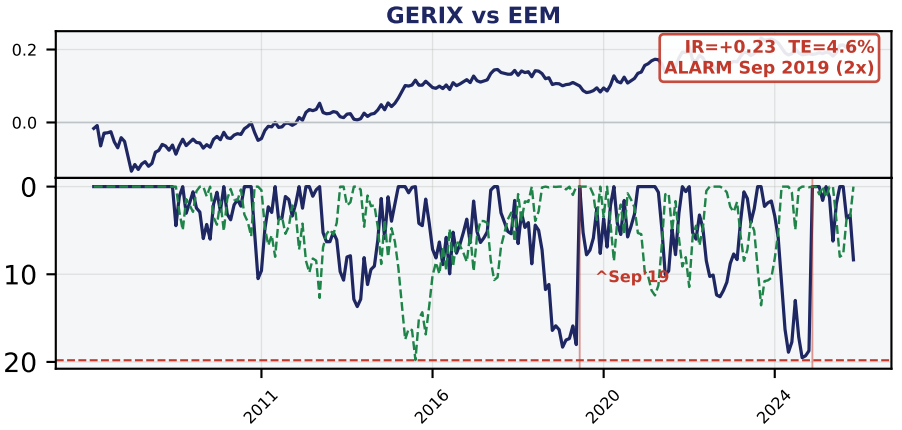


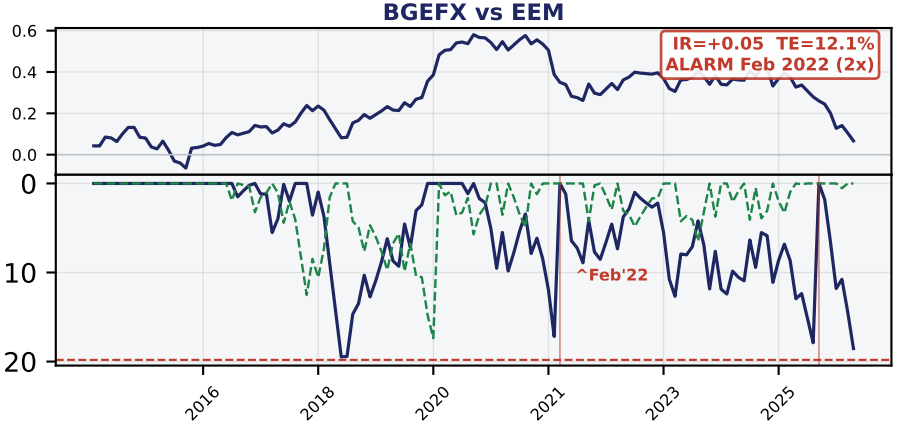


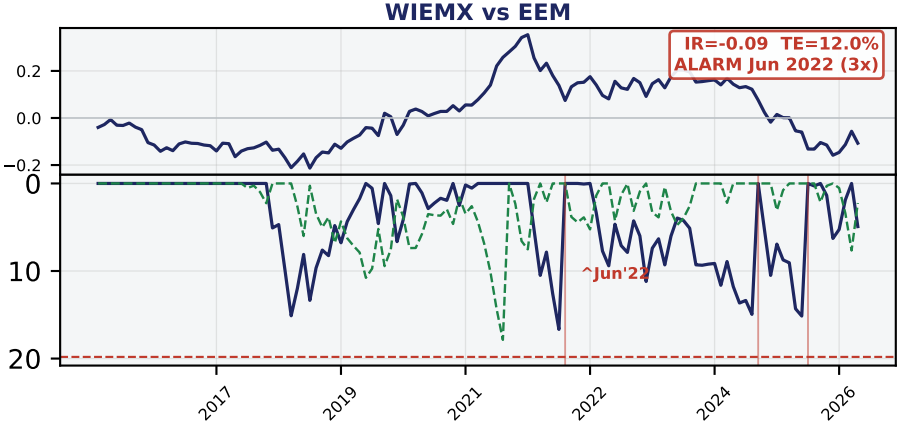


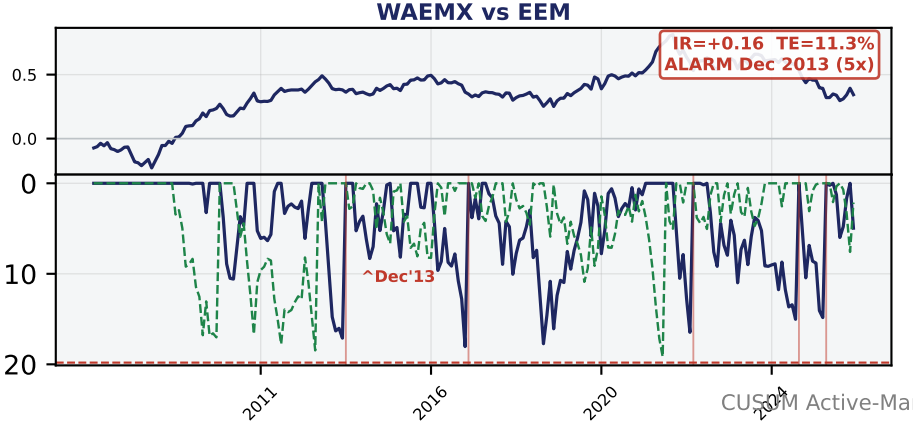


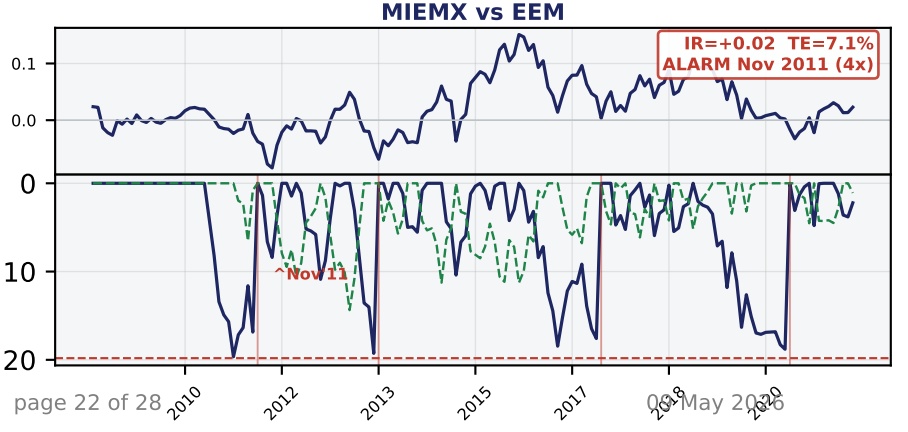
Individual Fund CUSUM Charts — page 6 of 12



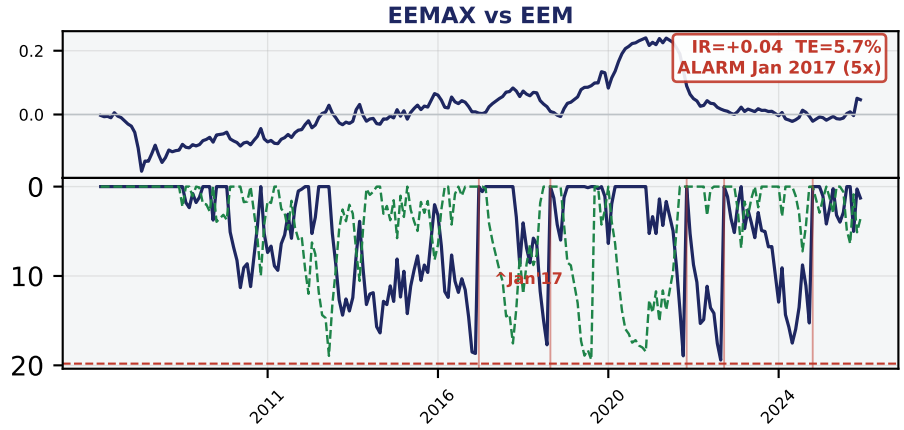


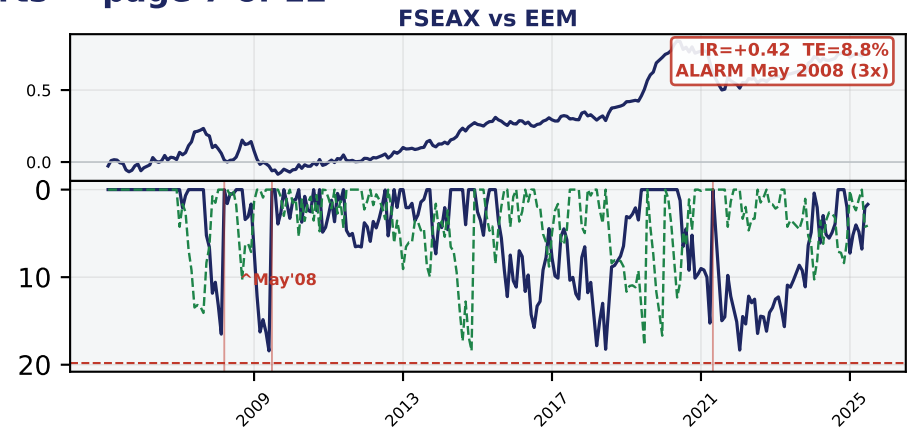


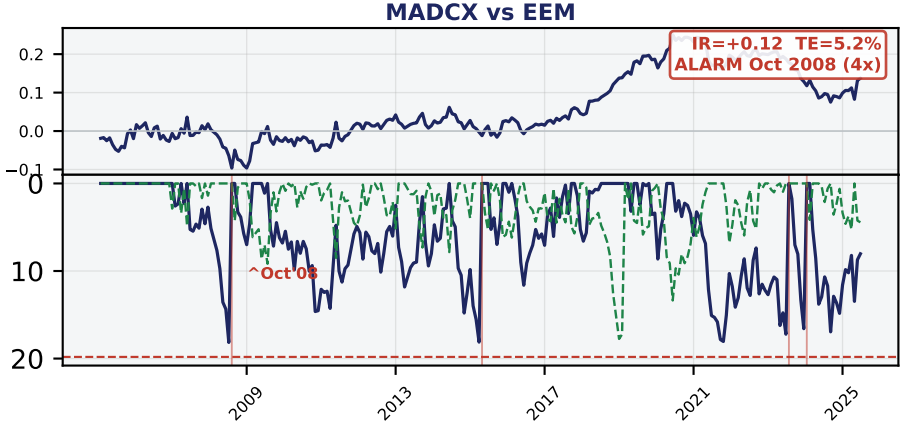


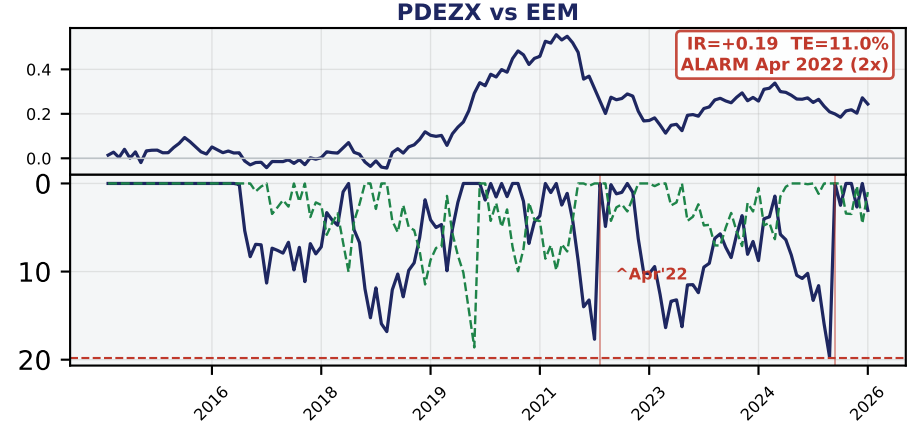


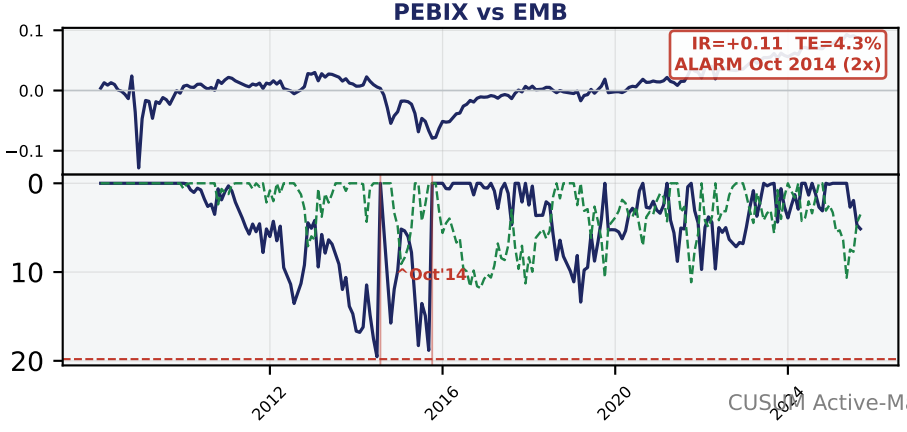
Individual Fund CUSUM Charts — page 7 of 12

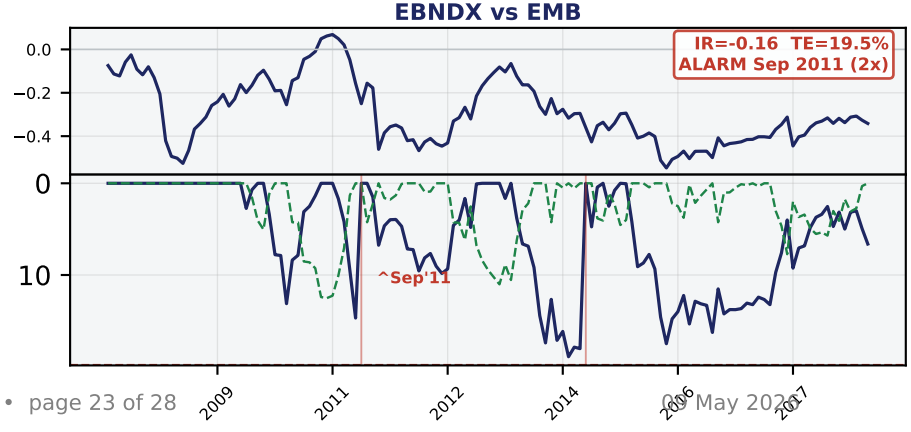




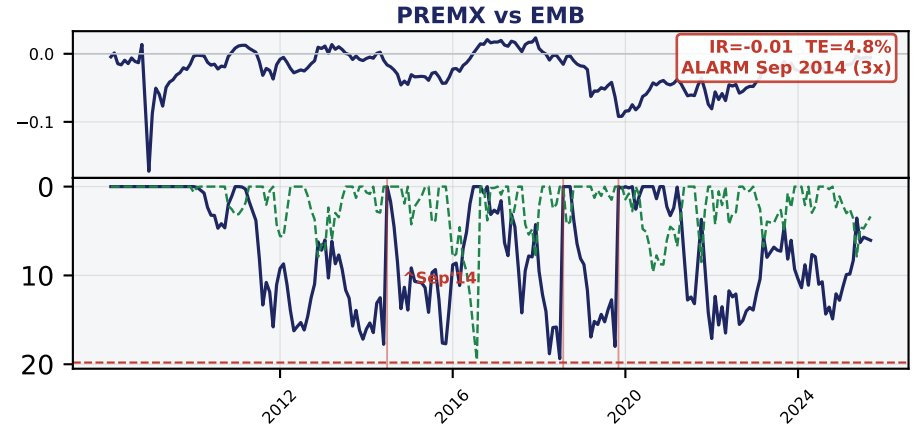


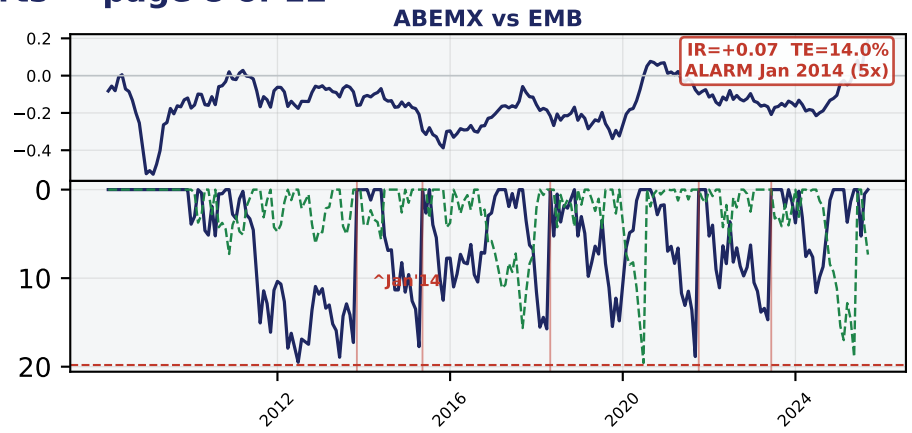


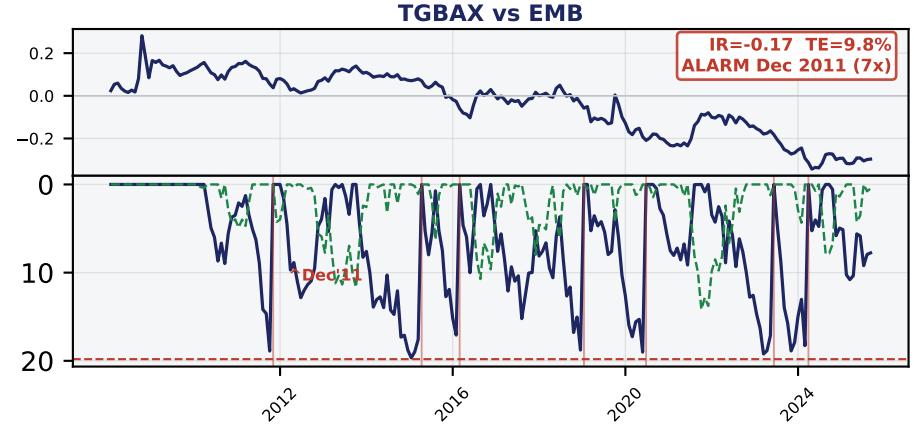


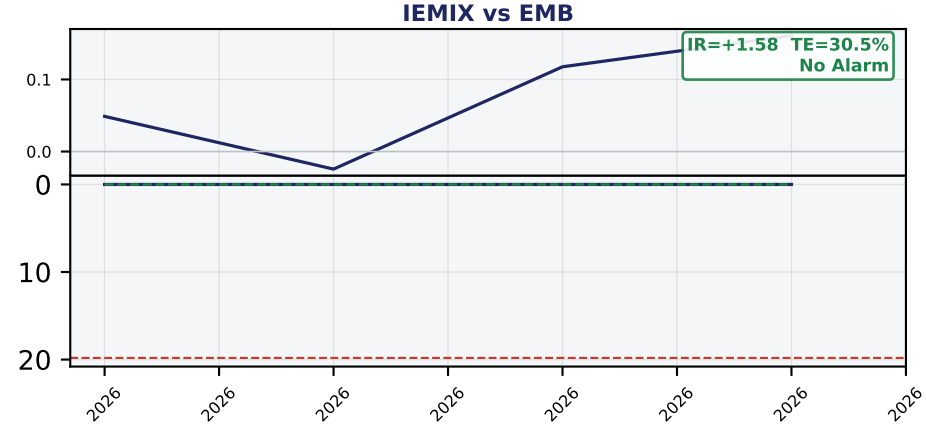


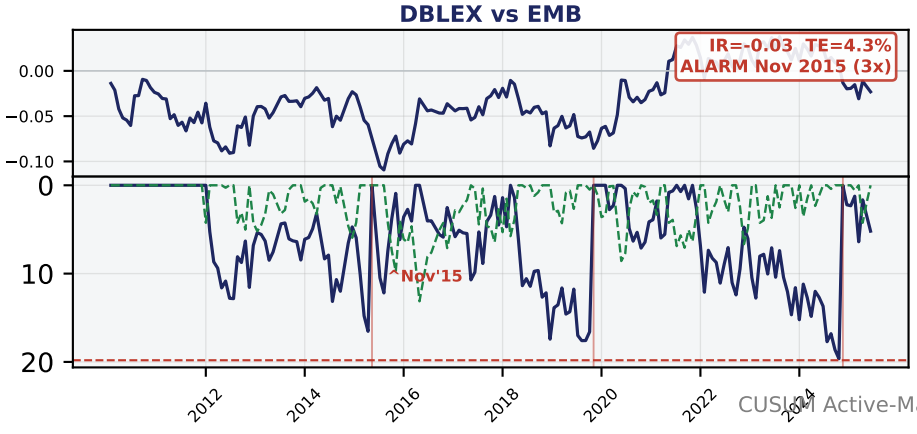
Individual Fund CUSUM Charts — page 8 of 12

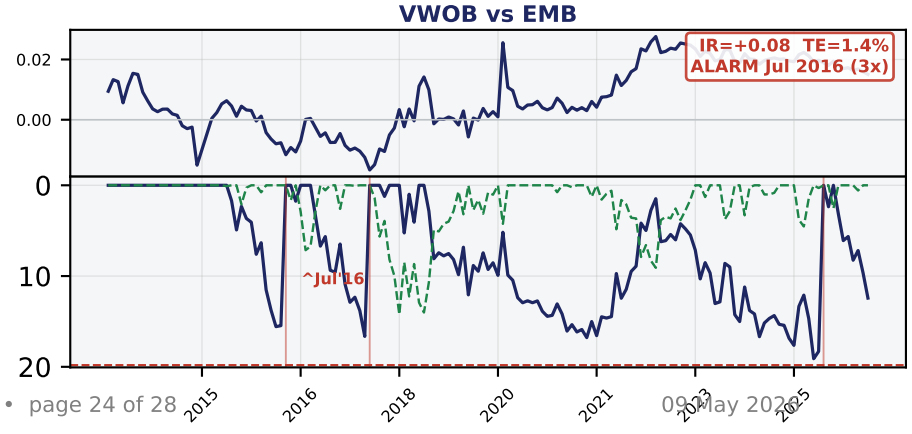




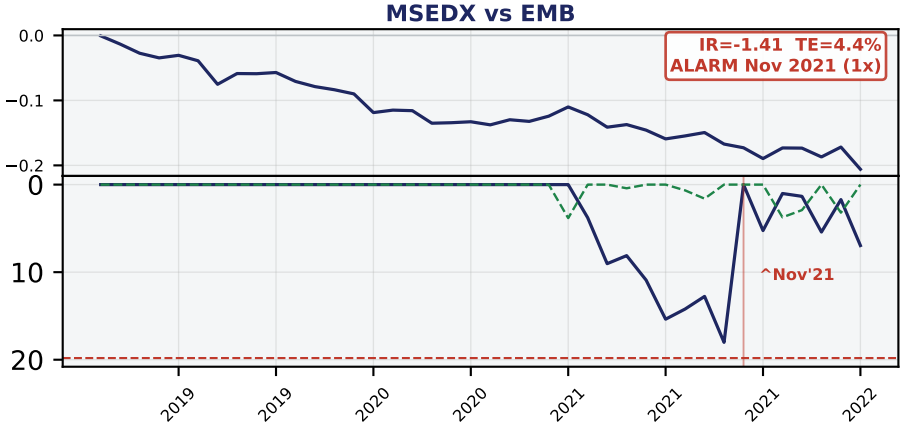


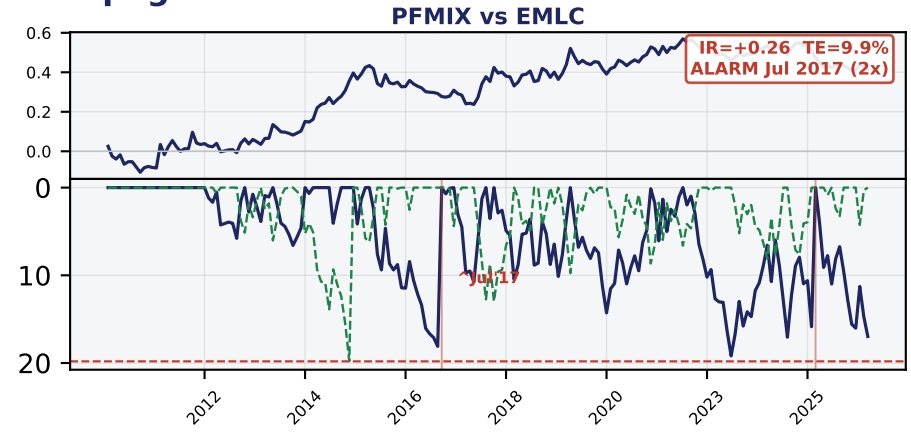


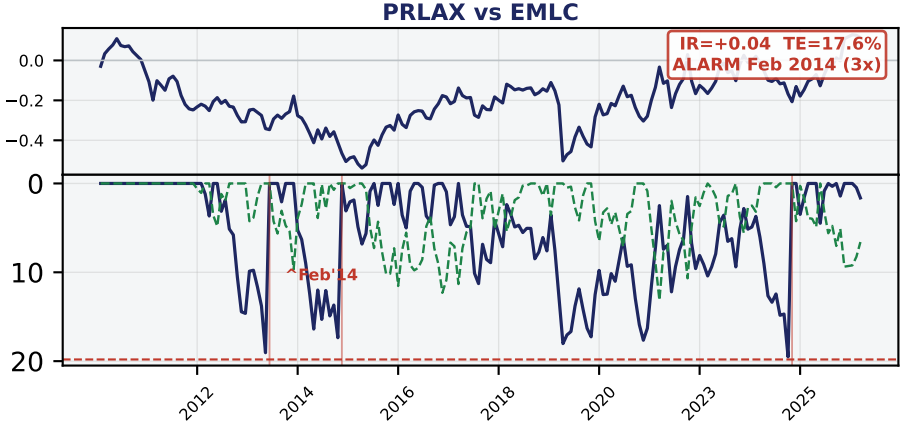


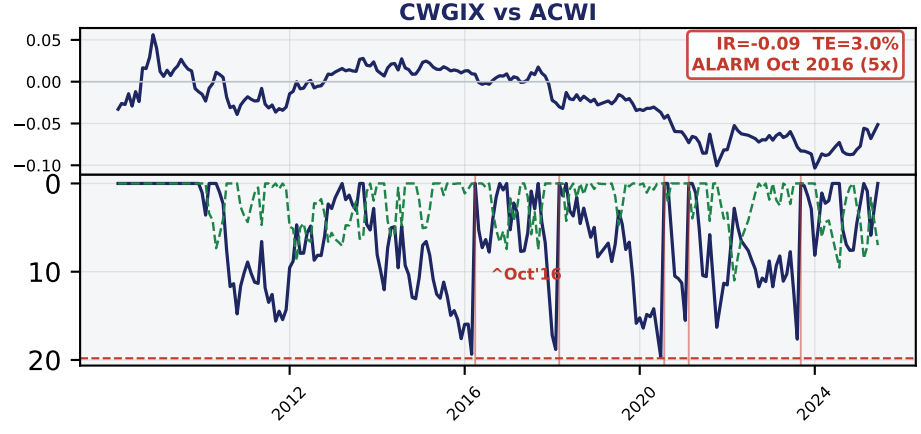


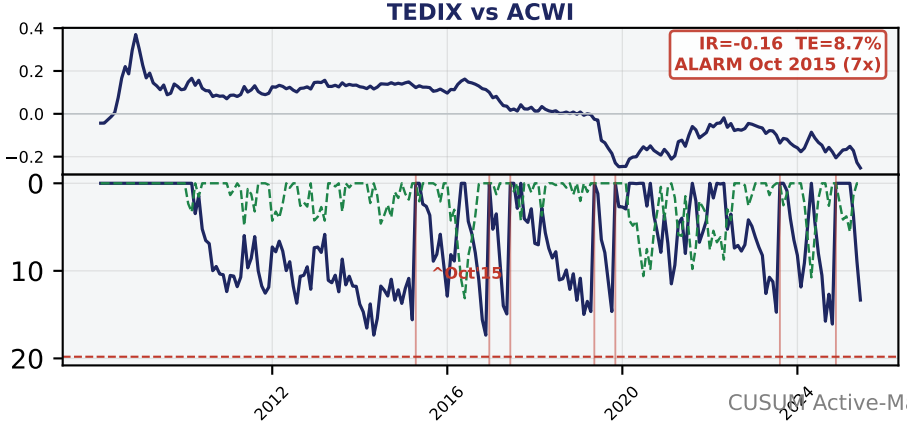
Individual Fund CUSUM Charts — page 9 of 12



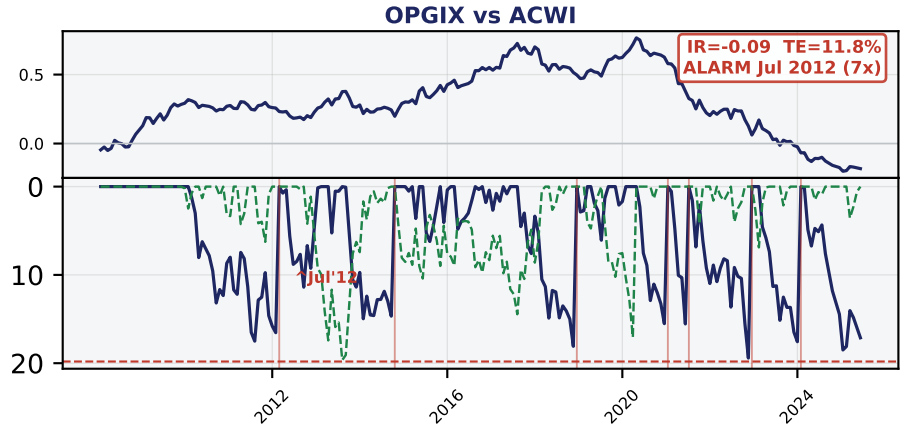


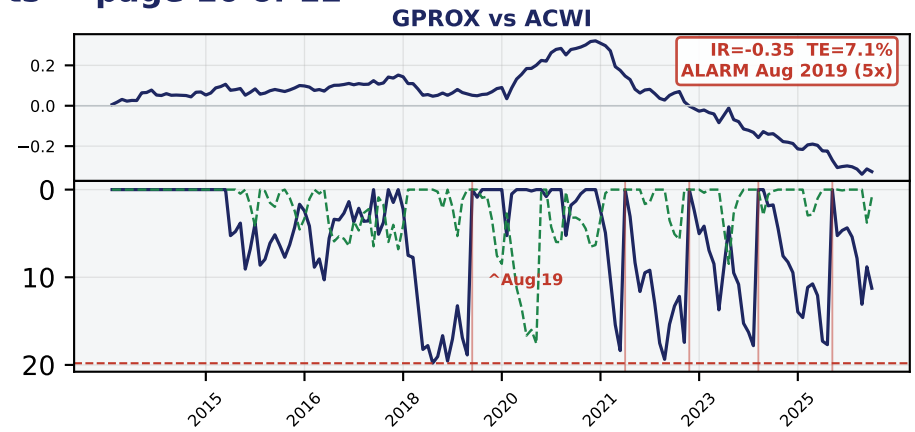


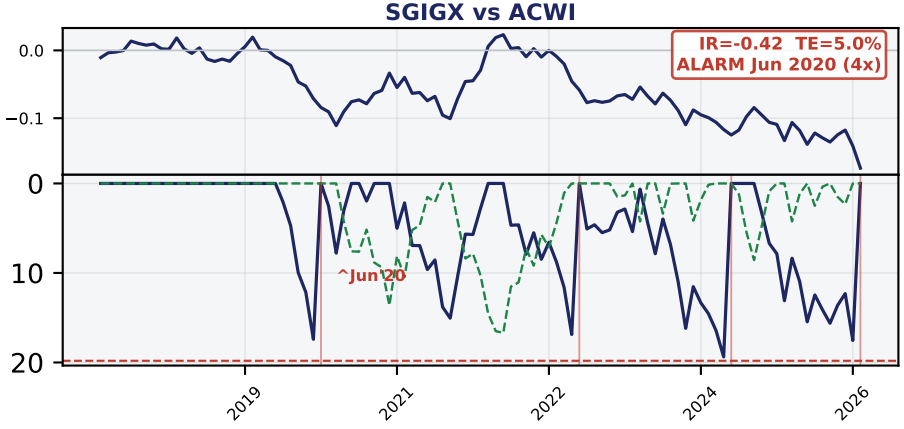


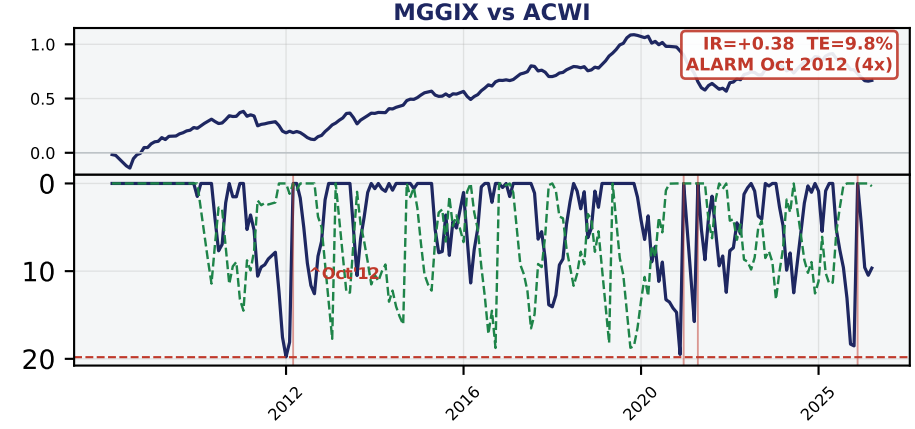


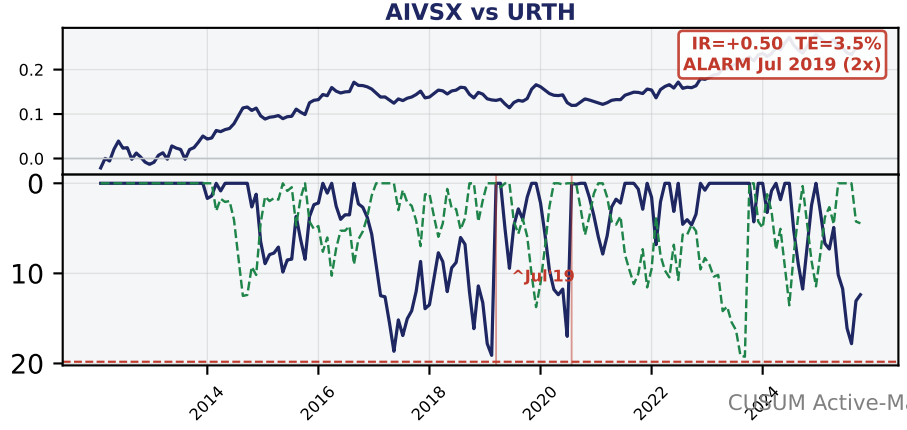
Individual Fund CUSUM Charts — page 10 of 12

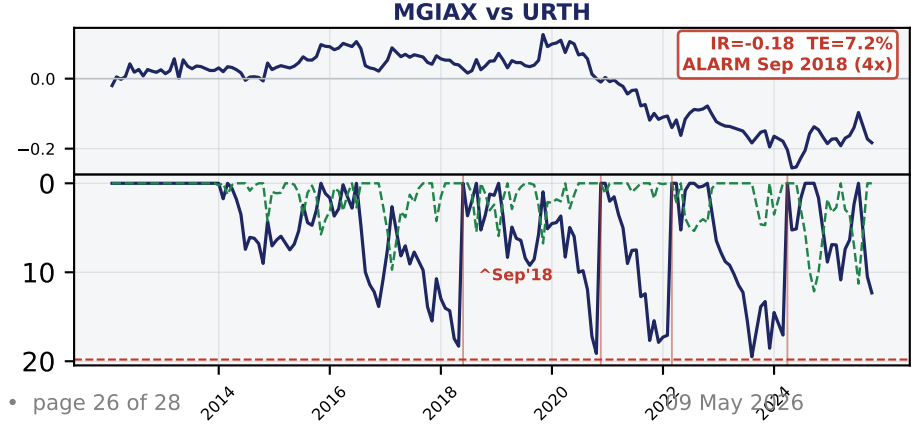




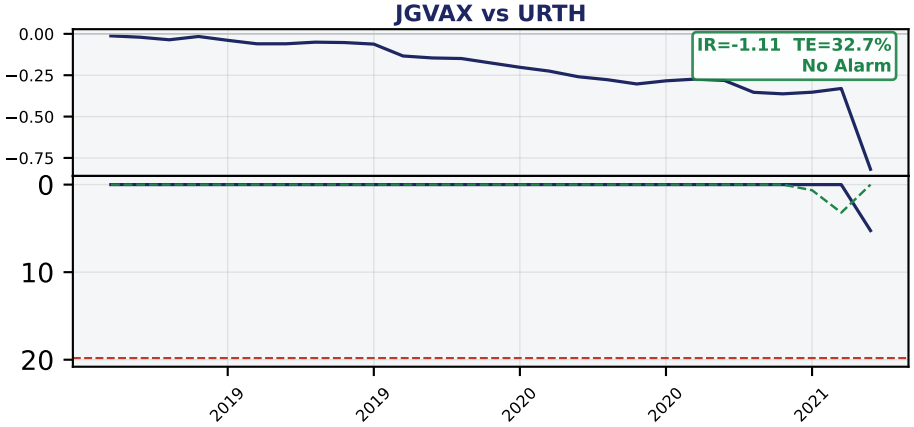


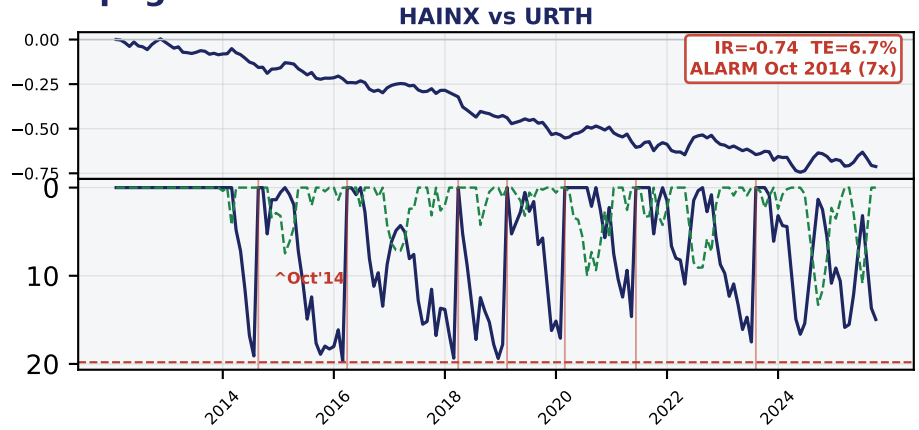


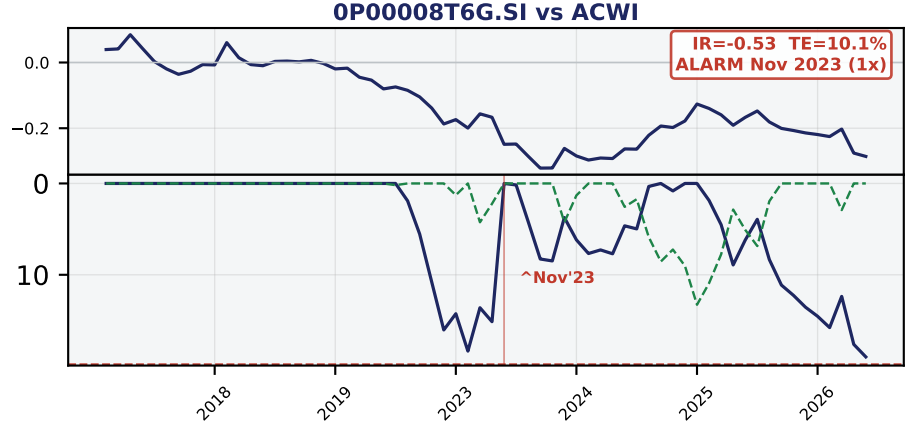
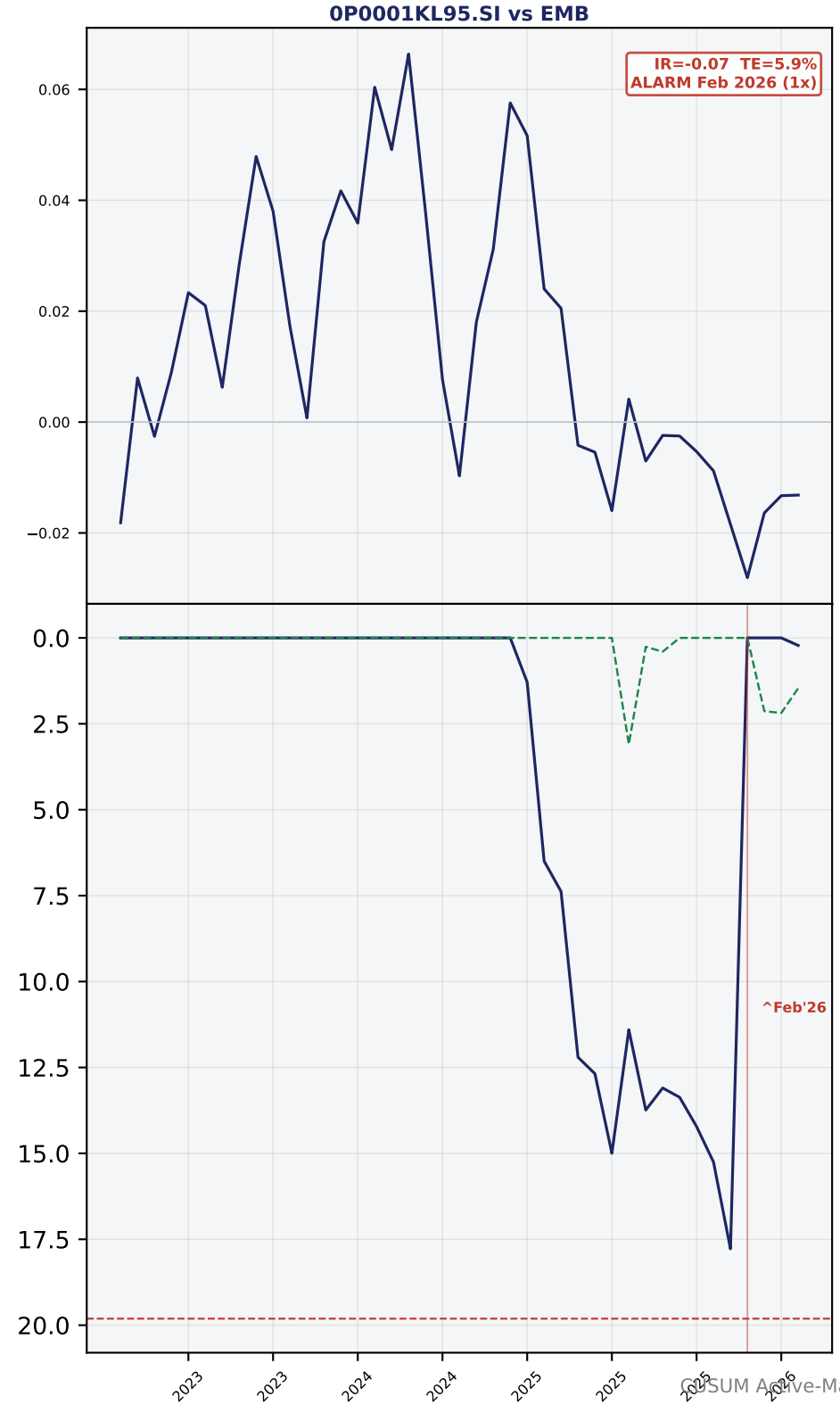
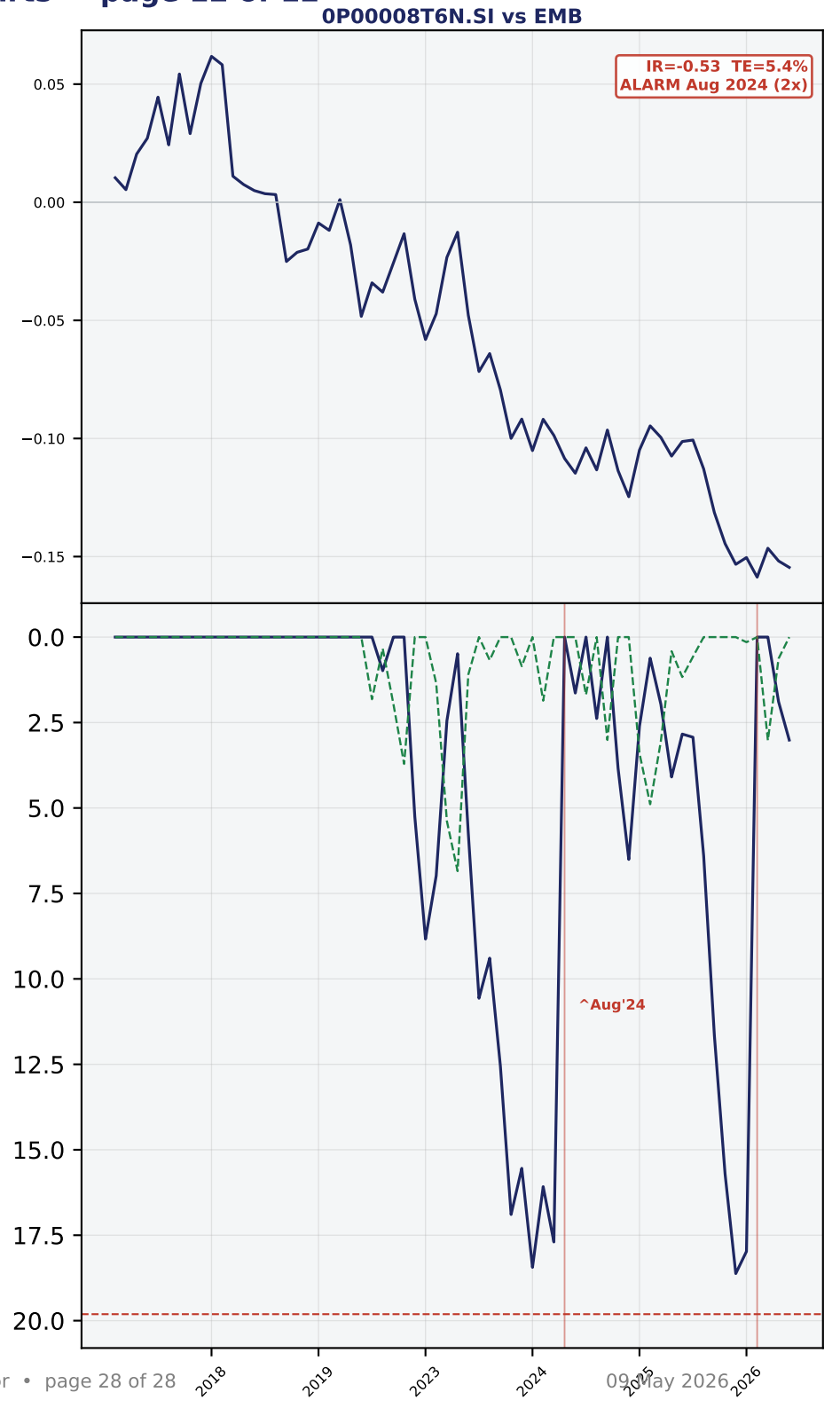




Individual Fund CUSUM Charts — page 11 of 12







Individual Fund CUSUM Charts — page 12 of 12